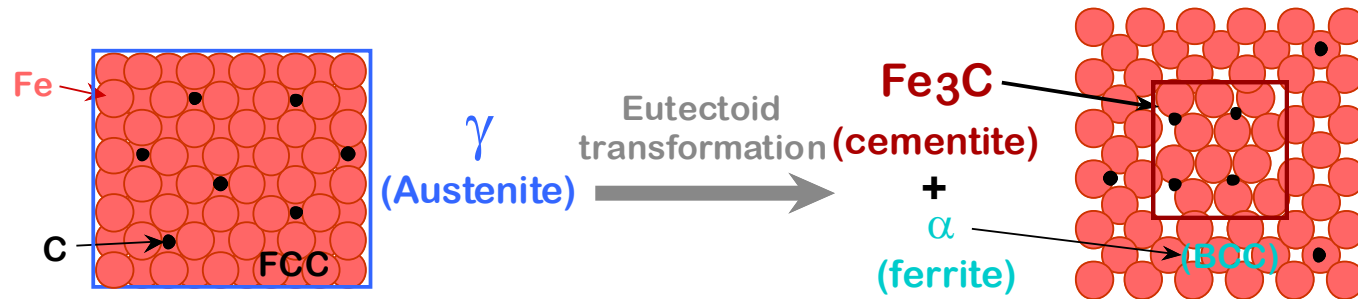


CHAPTER 10: PHASE TRANSFORMATIONS

ISSUES TO ADDRESS...

- Transforming one phase into another takes time.



- How does the rate of transformation depend on time and T?
- How can we slow down the transformation so that we can engineer non-equilibrium structures?
- Are the mechanical properties of non-equilibrium structures better?

Phase Transformations

BASIC CONCEPTS

A variety of **phase transformations** are important in the processing of materials, and usually they involve some alteration of the microstructure. For purposes of this discussion, these transformations are divided into three classifications.

In one group are simple diffusion-dependent transformations in which there is no change in either the number or composition of the phases present. These include solidification of a pure metal and allotropic transformations.

In another type of diffusion-dependent transformation, there is some alteration in phase compositions and often in the number of phases present; the final microstructure ordinarily consists of two phases.

The third kind of transformation is diffusionless, wherein a metastable phase is produced.

Nucleation and Growth

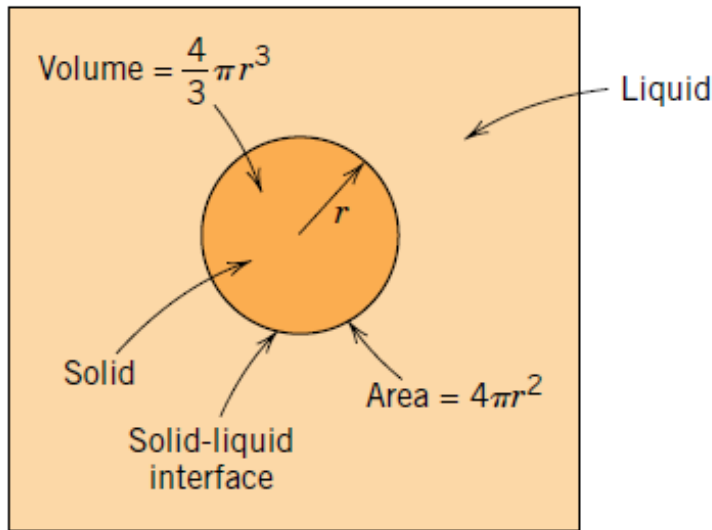
The progress of a phase transformation may be broken down into two distinct stages: **nucleation** and **growth**. Nucleation involves the appearance of very small particles, or nuclei of the new phase (often consisting of only a few hundred atoms), which are capable of growing. During the growth stage these nuclei increase in size, which results in the disappearance of some (or all) of the parent phase.

Nucleation

There are two types of nucleation: *homogeneous* and *heterogeneous*. The distinction between them is made according to the **site at which nucleating events occur**. For the homogeneous type, nuclei of the new phase form uniformly throughout the parent phase, whereas for the heterogeneous type, nuclei form preferentially at structural inhomogeneities, such as container surfaces, insoluble impurities, grain boundaries, dislocations, and so on.

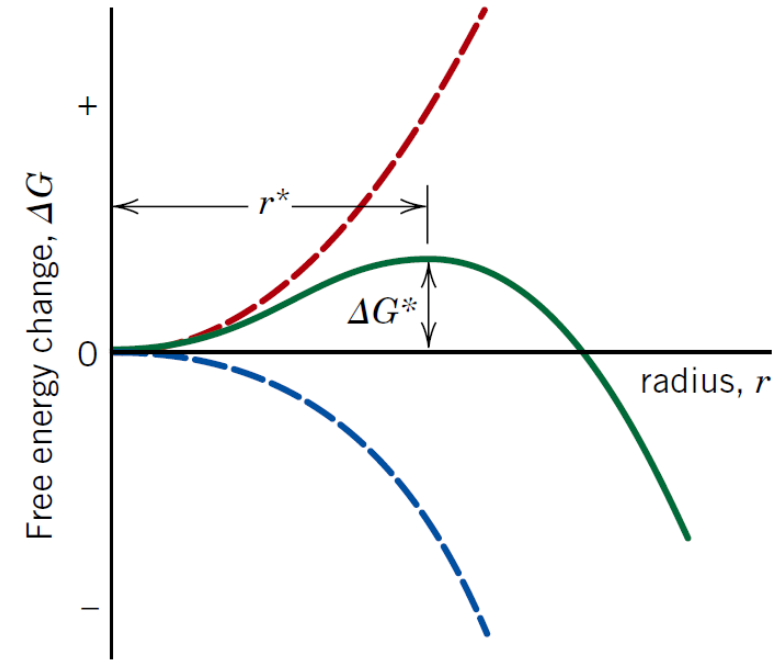
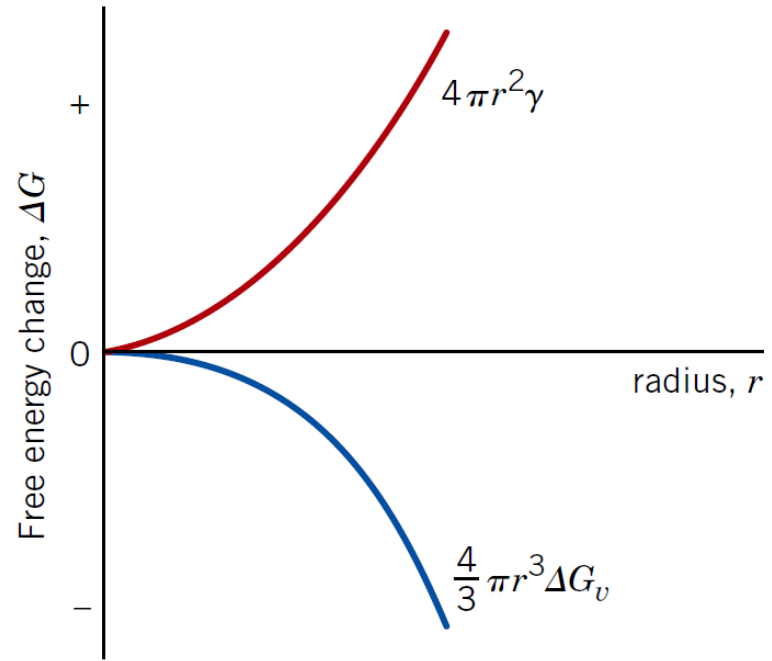
A discussion of the theory of nucleation involves a thermodynamic parameter called **free energy** (or *Gibbs free energy*), G . In brief, free energy is a function of other thermodynamic parameters

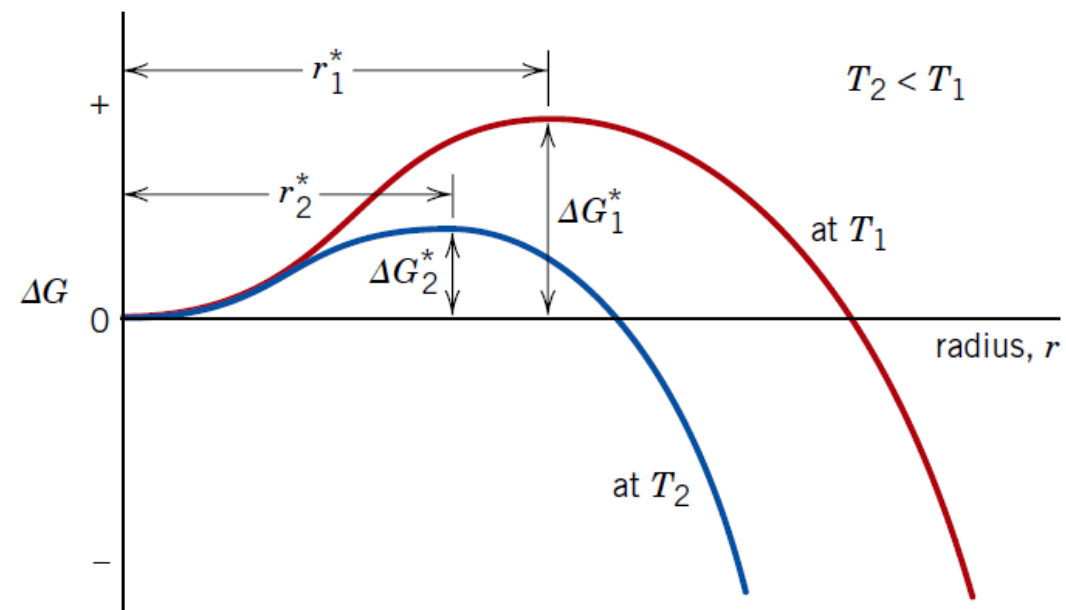
A transformation will occur spontaneously only when G has a negative value.



$$\Delta G = \frac{4}{3}\pi r^3 \Delta G_v + 4\pi r^2 \gamma$$

$$\Delta G = \frac{4}{3}\pi r^3 \Delta G_v + 4\pi r^2 \gamma$$





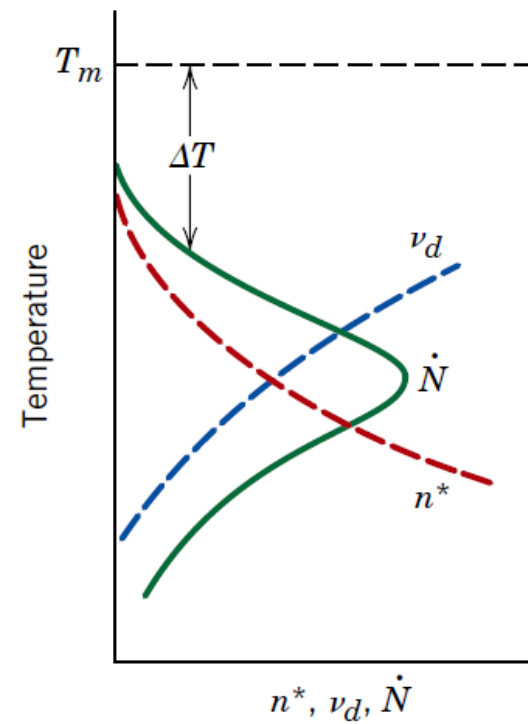
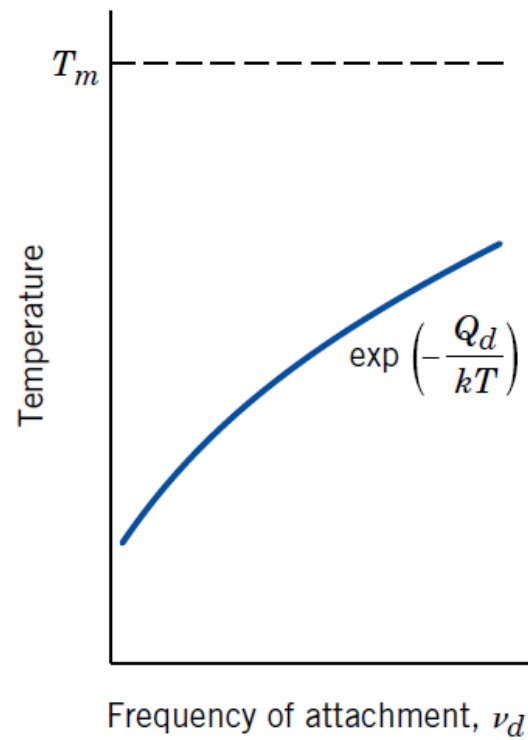
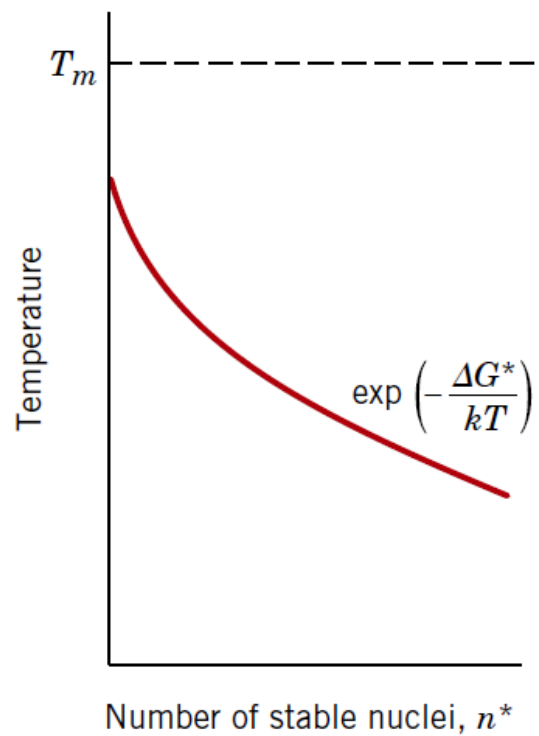


Table 10.1 Degree of Supercooling (ΔT) Values
(Homogeneous Nucleation) for
Several Metals

<i>Metal</i>	ΔT ($^{\circ}\text{C}$)
Antimony	135
Germanium	227
Silver	227
Gold	230
Copper	236
Iron	295
Nickel	319
Cobalt	330
Palladium	332

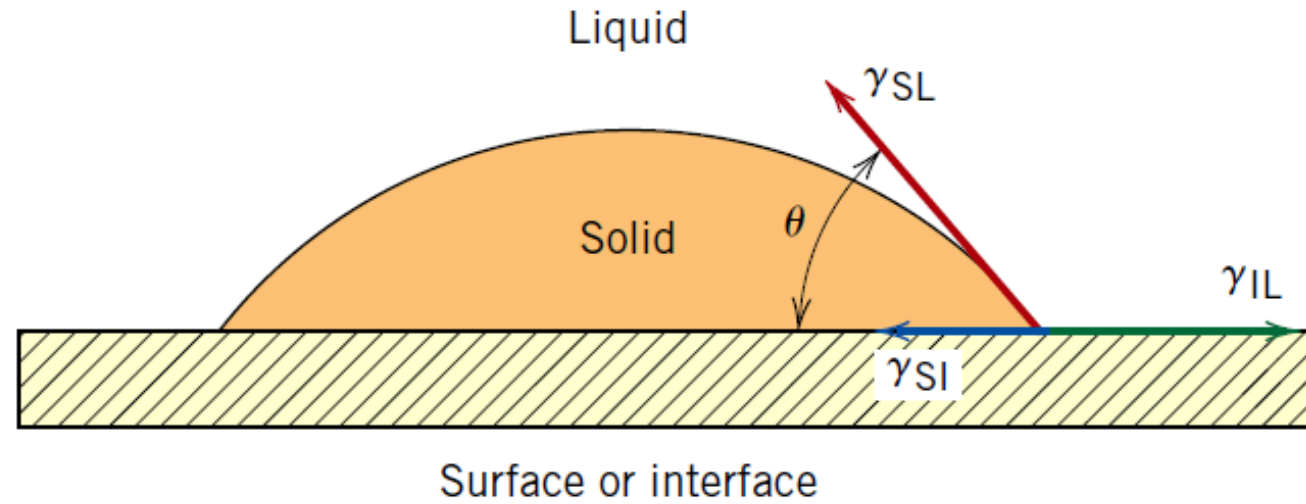
Source: D. Turnbull and R. E. Cech, “Microscopic Observation of the Solidification of Small Metal Droplets,” *J. Appl. Phys.*, **21**, 808 (1950).

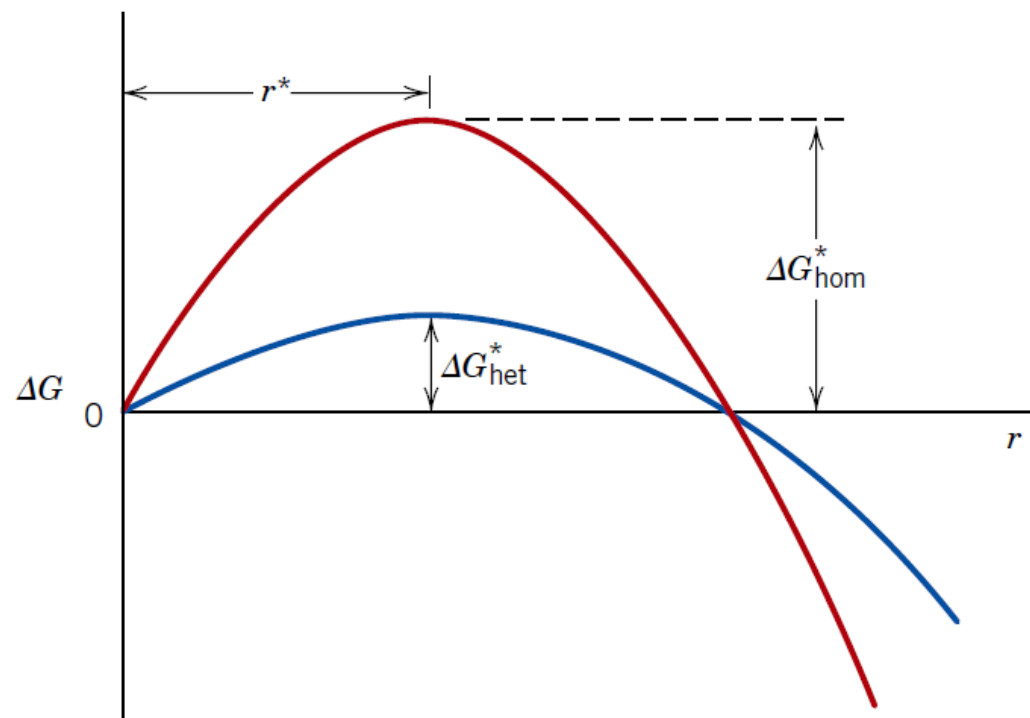
(a) For the solidification of pure gold, calculate the critical radius r^* and the activation free energy ΔG^* if nucleation is homogeneous. Values for the latent heat of fusion and surface free energy are $-1.16 \times 10^9 \text{ J/m}^3$ and 0.132 J/m^2 , respectively. Use the supercooling value found in Table 10.1.

(b) Now calculate the number of atoms found in a nucleus of critical size. Assume a lattice parameter of 0.413 nm for solid gold at its melting temperature.

Heterogeneous Nucleation

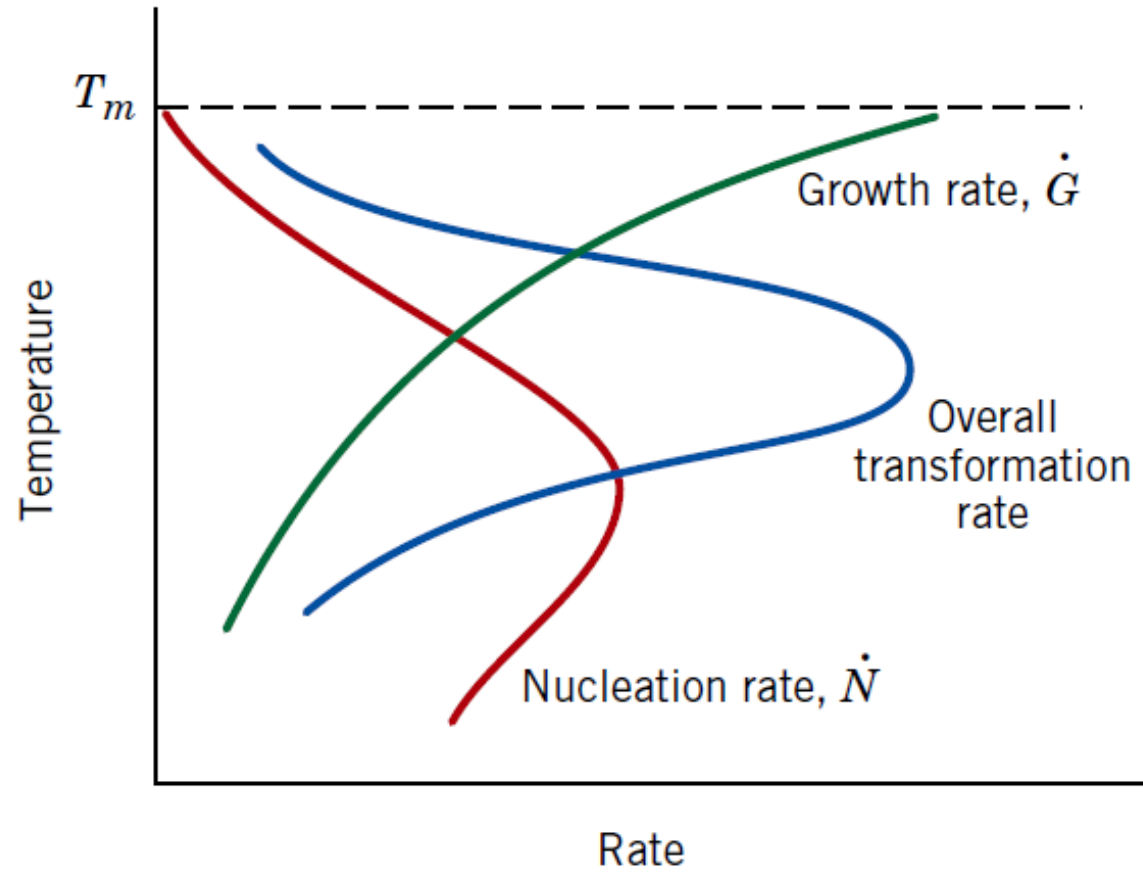
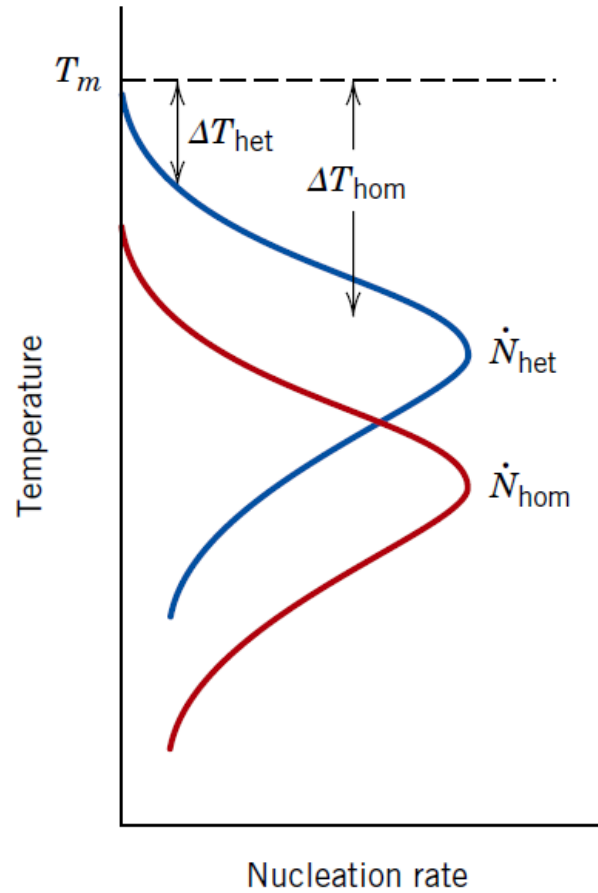
$$\gamma_{IL} = \gamma_{SI} + \gamma_{SL} \cos \theta$$

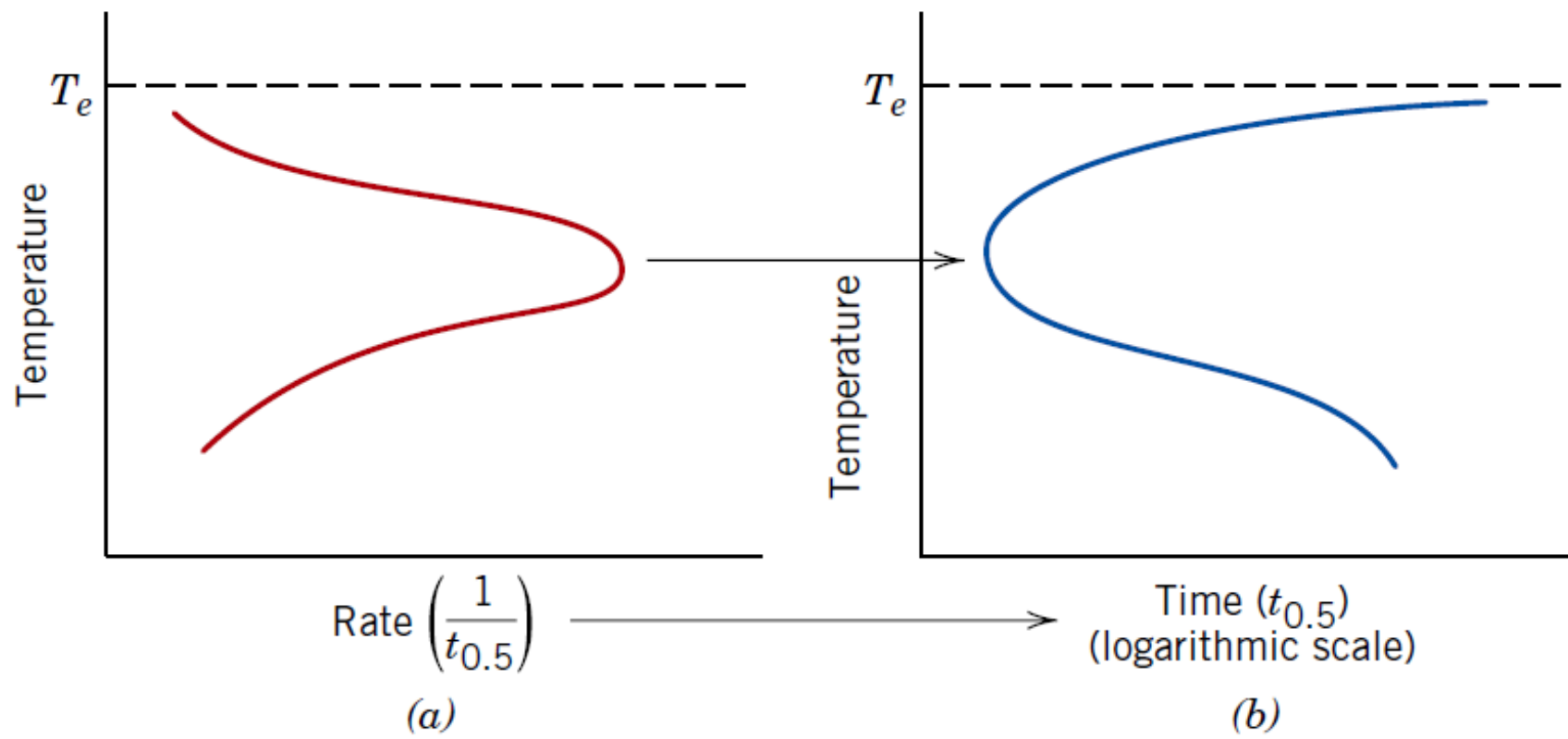




Growth

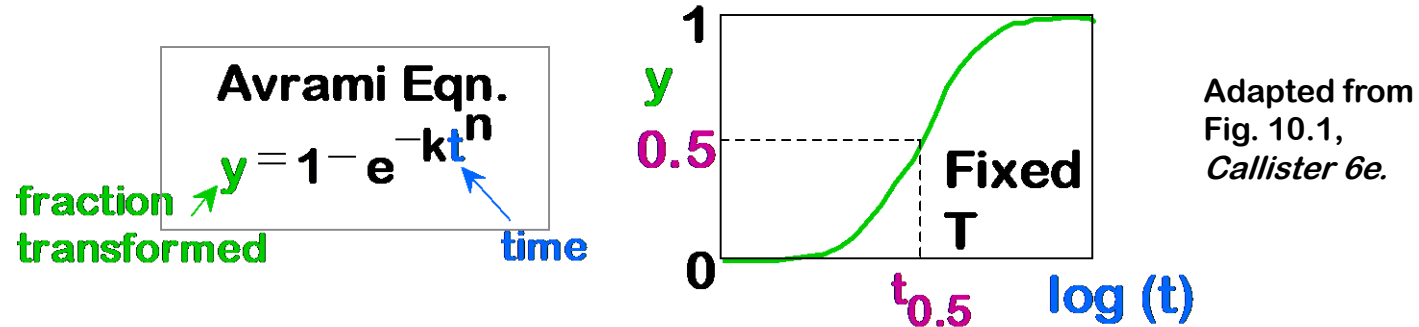
$$\dot{G} = C \exp \left(- \frac{Q}{kT} \right)$$





FRACTION OF TRANSFORMATION

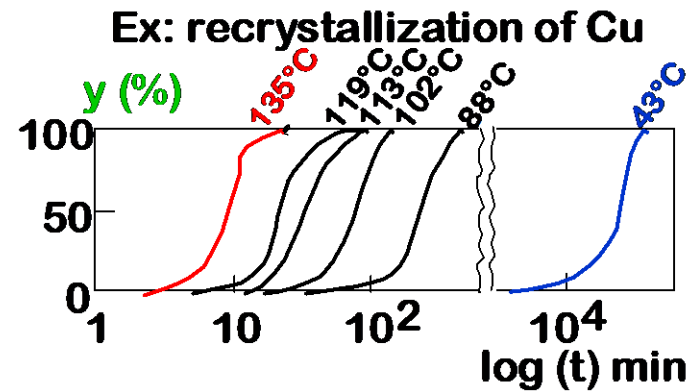
- **Fraction transformed** depends on time.



- Transformation rate depends on T .

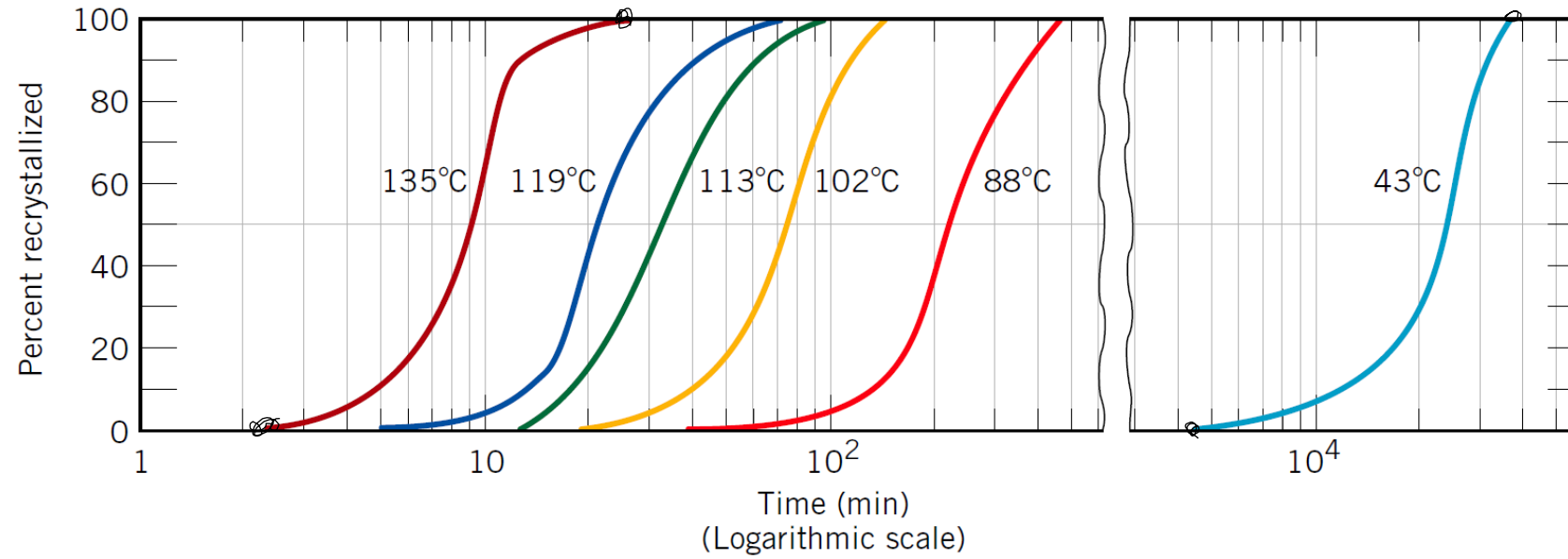
activation energy

$$r = \frac{1}{t_{0.5}} = Ae^{-Q/RT}$$



- r often small: equil not possible!

$$y = 1 - \exp(-kt^n)$$



$$\text{rate} = \frac{1}{t_{0.5}}$$

Rate of Recrystallization Computation

It is known that the kinetics of recrystallization follows the Avrami equation and that the value of n is 3.1. If the fraction crystallized is 0.3 after 20 min, determine the rate of recrystallization.

$$\text{rate} = \frac{1}{t_{0.5}} \quad Y = 1 - \exp(-K t^n)$$

$$\exp(-K t^n) = 1 - Y \rightarrow -K t^n = \ln(1 - Y)$$

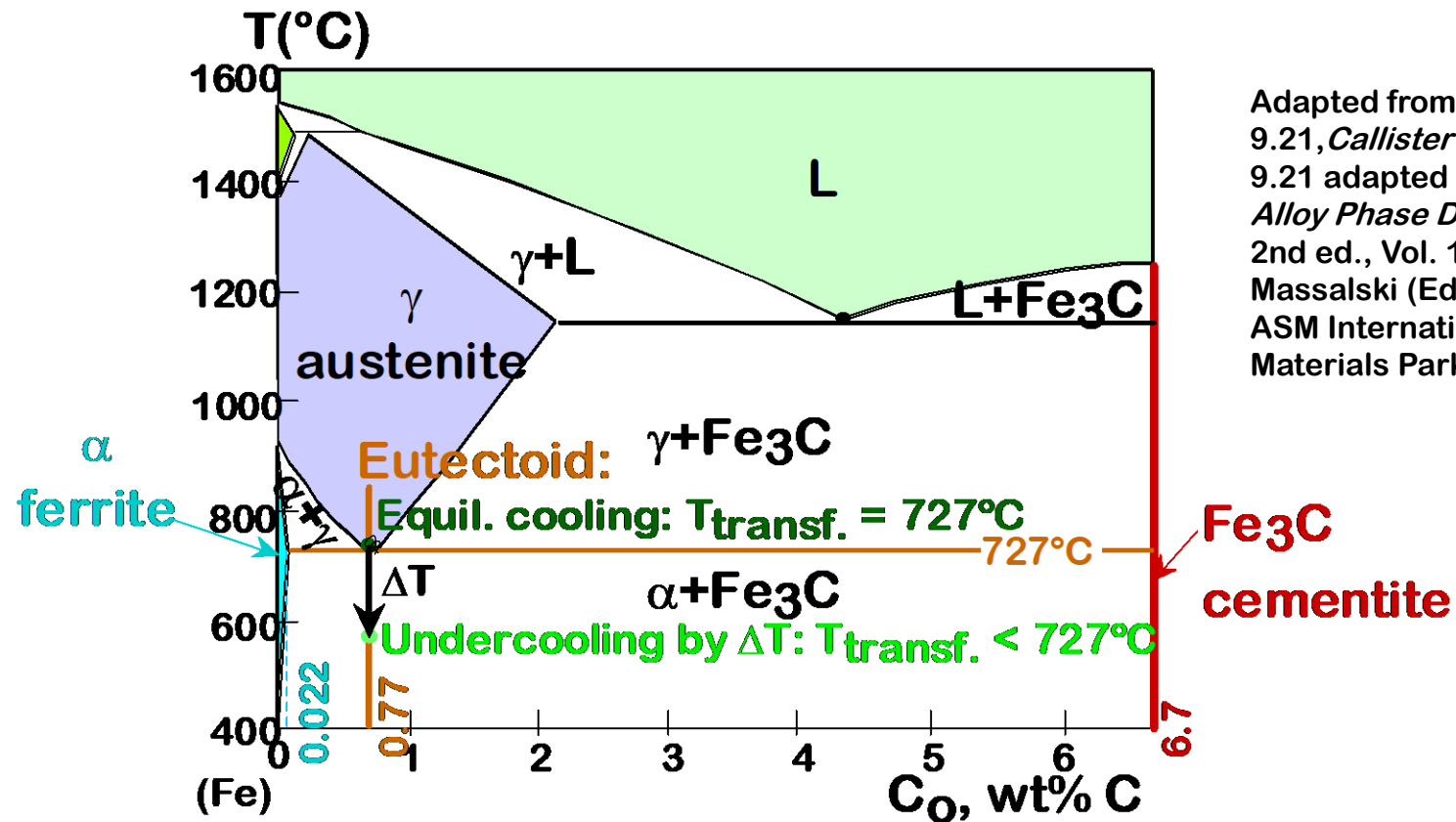
$$K = - \frac{\ln(1 - 0.3)}{(20 \text{ min})^{3.1}} = 3.3 \times 10^{-5}$$

$$t^n = - \frac{\ln(1 - Y)}{K} \rightarrow t = \left[- \frac{\ln(1 - Y)}{K} \right]^{1/n}$$

$$Y = 0.3, n = 3.1, K = 3.3 \times 10^{-5} \rightarrow \begin{cases} t_{0.5} = 24.8 \text{ min} \\ \text{rate} = \frac{1}{t_{0.5}} = 4 \times 10^{-2} [\text{min}]^{-1} \end{cases}$$

TRANSFORMATIONS & UNDERCOOLING

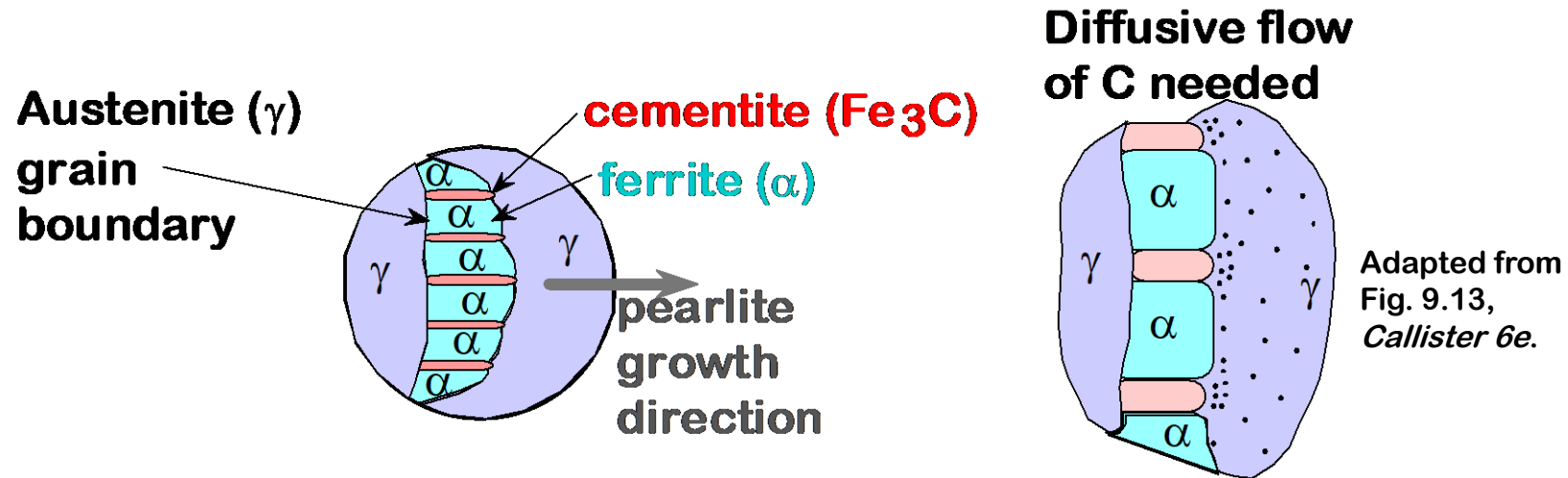
- **Eutectoid transf. (Fe-C System):** $\gamma \Rightarrow \alpha + \text{Fe}_3\text{C}$
- **Can make it occur at:**
 - ...727°C (cool it slowly)
 - ...below 727°C (“undercool” it!)



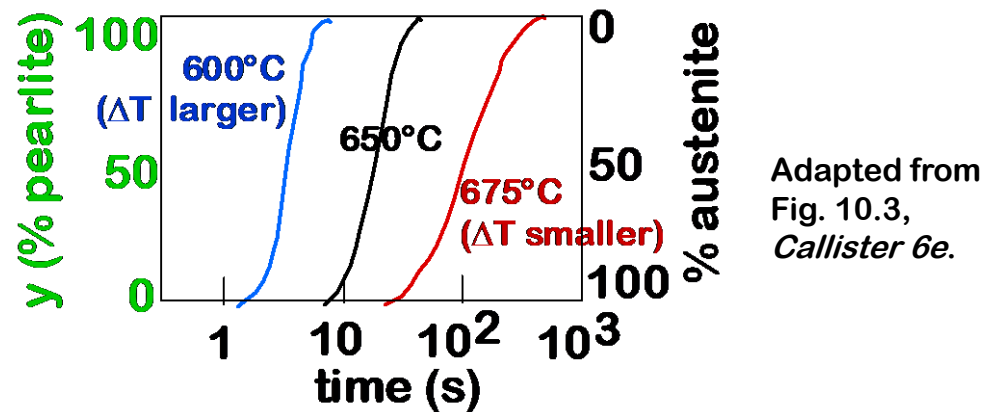
Adapted from Fig. 9.21, *Callister 6e*. (Fig. 9.21 adapted from *Binary Alloy Phase Diagrams*, 2nd ed., Vol. 1, T.B. Massalski (Ed.-in-Chief), ASM International, Materials Park, OH, 1990.)

EUTECTOID TRANSFORMATION RATE $\sim \Delta T$

- Growth of pearlite from austenite:

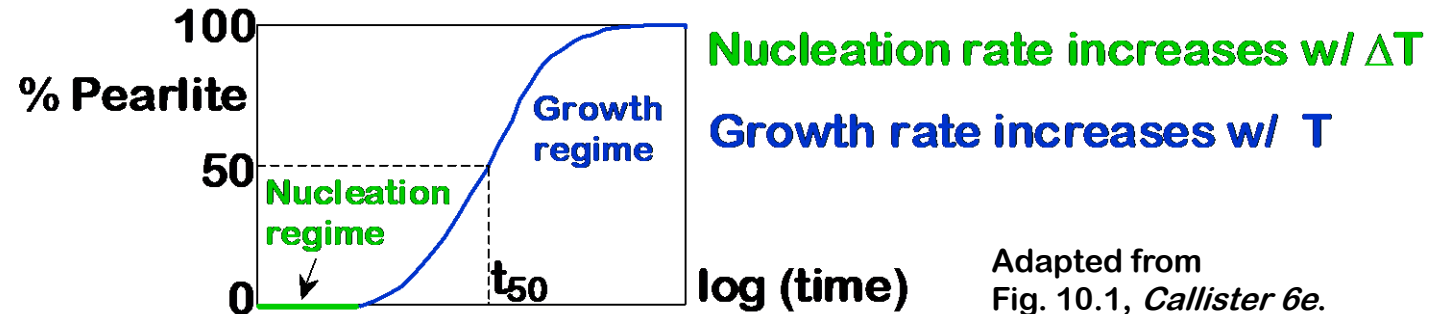


- Reaction rate increases with ΔT .

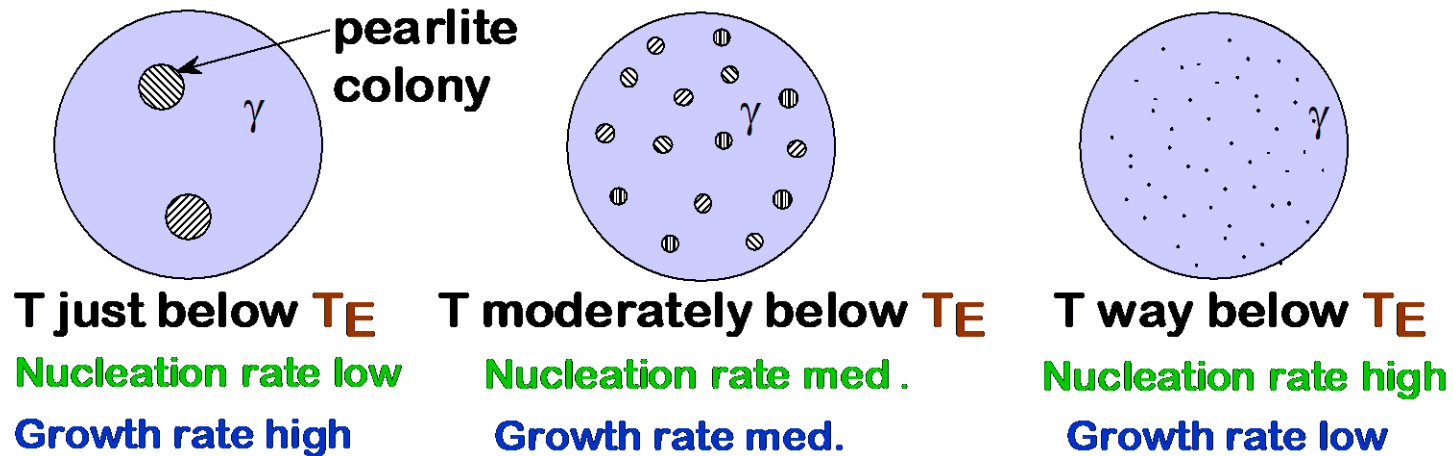


NUCLEATION AND GROWTH

- Reaction rate is a result of nucleation and growth of crystals.

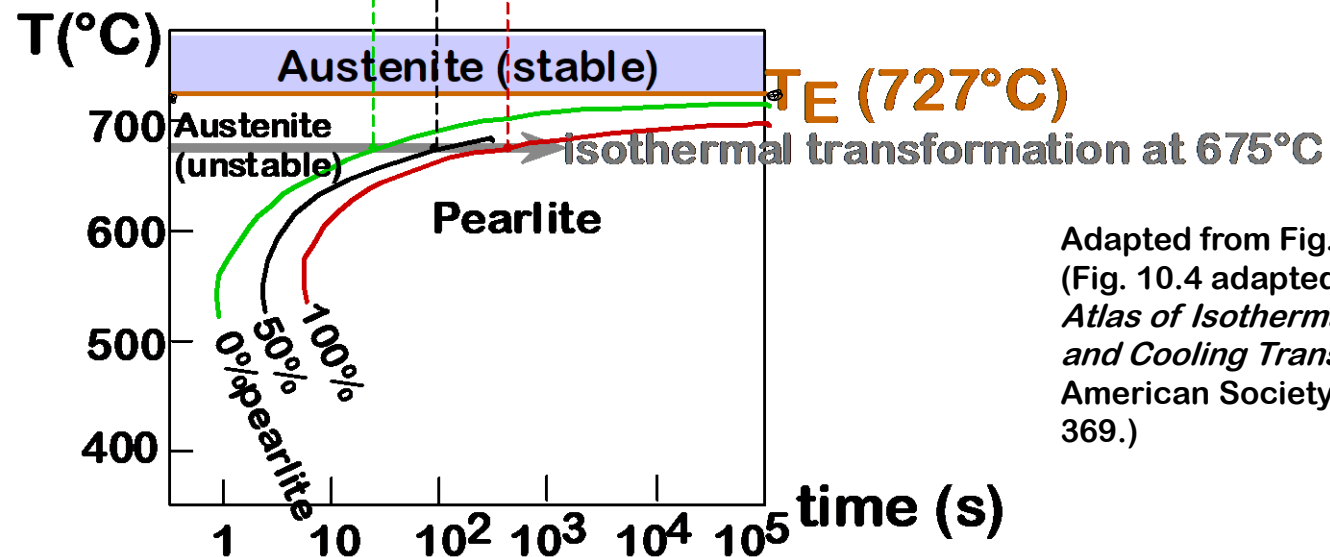
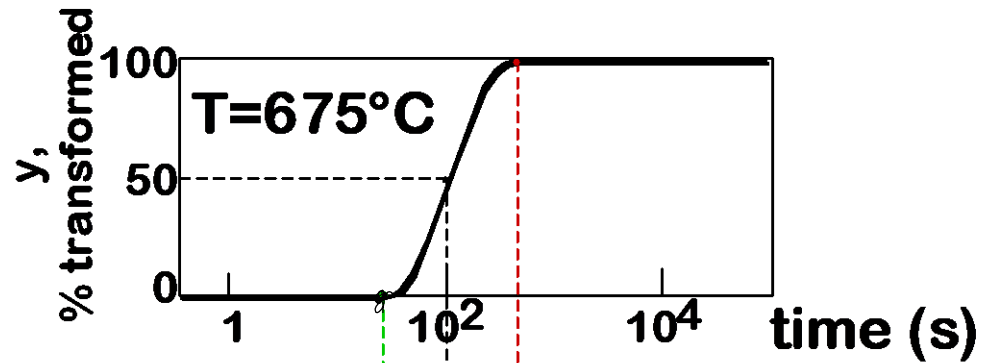


- Examples:



ISOTHERMAL TRANSFORMATION DIAGRAMS

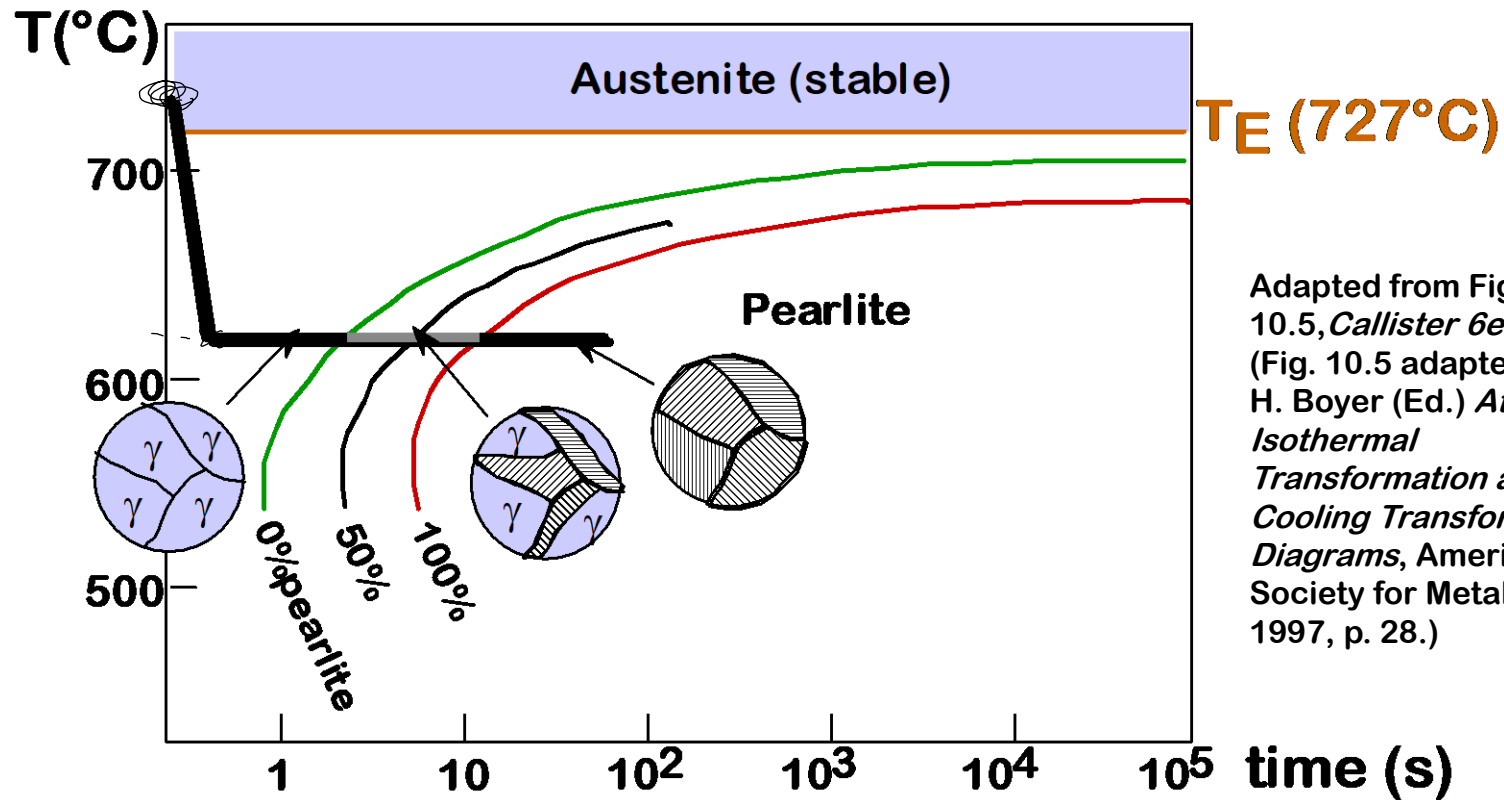
- Fe-C system, $C_0 = 0.77\text{wt}\%\text{C}$
- Transformation at $T = 675^\circ\text{C}$.



Adapted from Fig. 10.4, *Callister 6e*.
(Fig. 10.4 adapted from H. Boyer (Ed.)
*Atlas of Isothermal Transformation
and Cooling Transformation Diagrams*,
American Society for Metals, 1977, p.
369.)

EX: COOLING HISTORY Fe-C SYSTEM

- Eutectoid composition, $C_0 = 0.77\text{wt}\%\text{C}$
- Begin at $T > 727^\circ\text{C}$
- Rapidly cool to 625°C and hold isothermally.



PEARLITE MORPHOLOGY

Two cases:

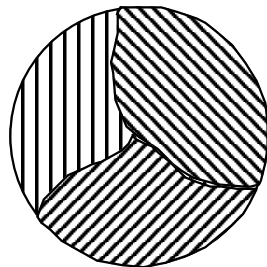
- T_{transf} just below T_E
 - Larger T : diffusion is faster
 - Pearlite is coarser.
- T_{transf} well below T_E
 - Smaller T : diffusion is slower
 - Pearlite is finer.



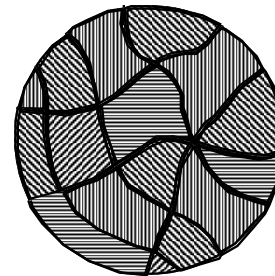
10 μm



Adapted from Fig. 10.6 (a) and (b), *Callister 6e*. (Fig. 10.6 from R.M. Ralls et al., *An Introduction to Materials Science and Engineering*, p. 361, John Wiley and Sons, Inc., New York, 1976.)



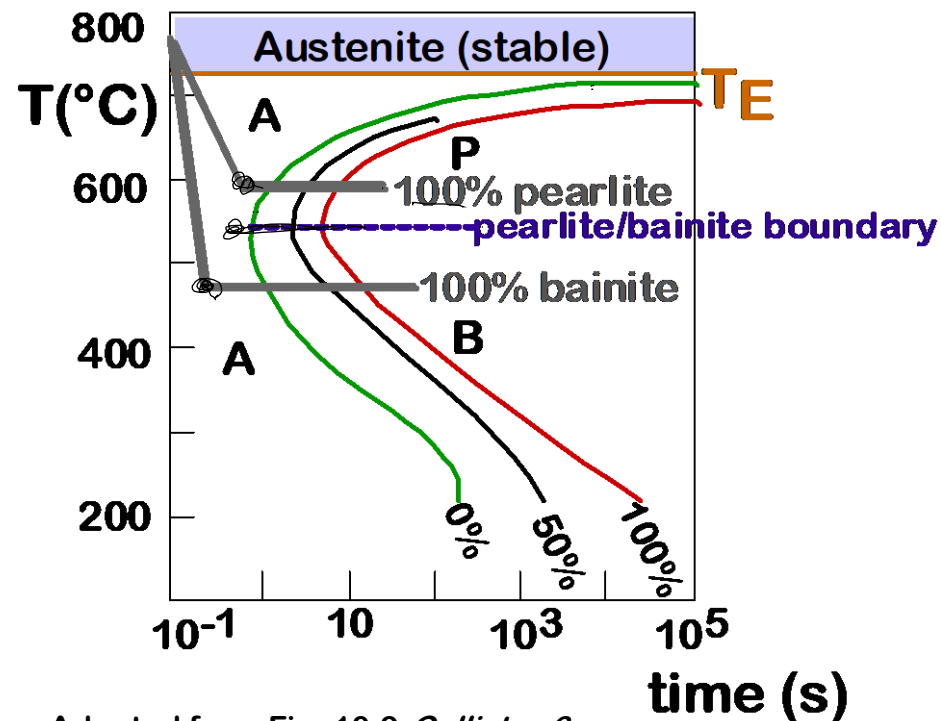
- Smaller ΔT :
colonies are
larger



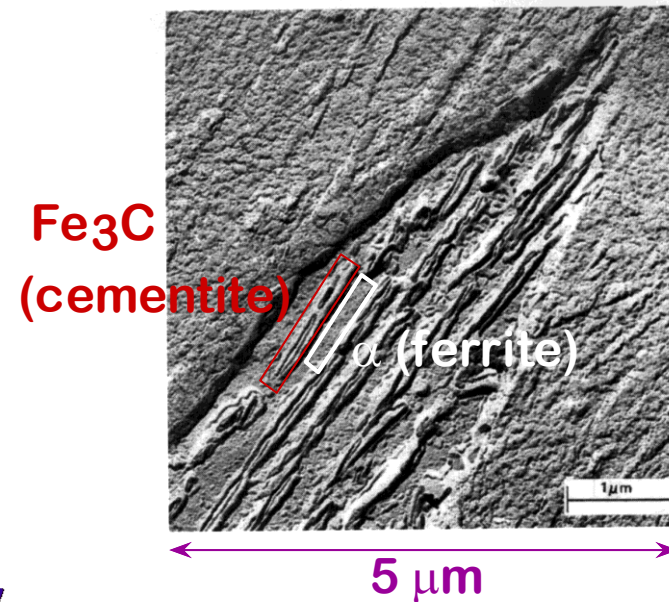
- Larger ΔT :
colonies are
smaller

NON-EQUIL TRANSFORMATION PRODUCTS: Fe-C

- **Bainite:**
 - α lathes (strips) with long rods of Fe_3C
 - diffusion controlled.
- Isothermal Transf. Diagram



Adapted from Fig. 10.9, *Callister 6e*.
(Fig. 10.9 adapted from H. Boyer (Ed.) *Atlas of Isothermal Transformation and Cooling Transformation Diagrams*, American Society for Metals, 1997, p. 28.)



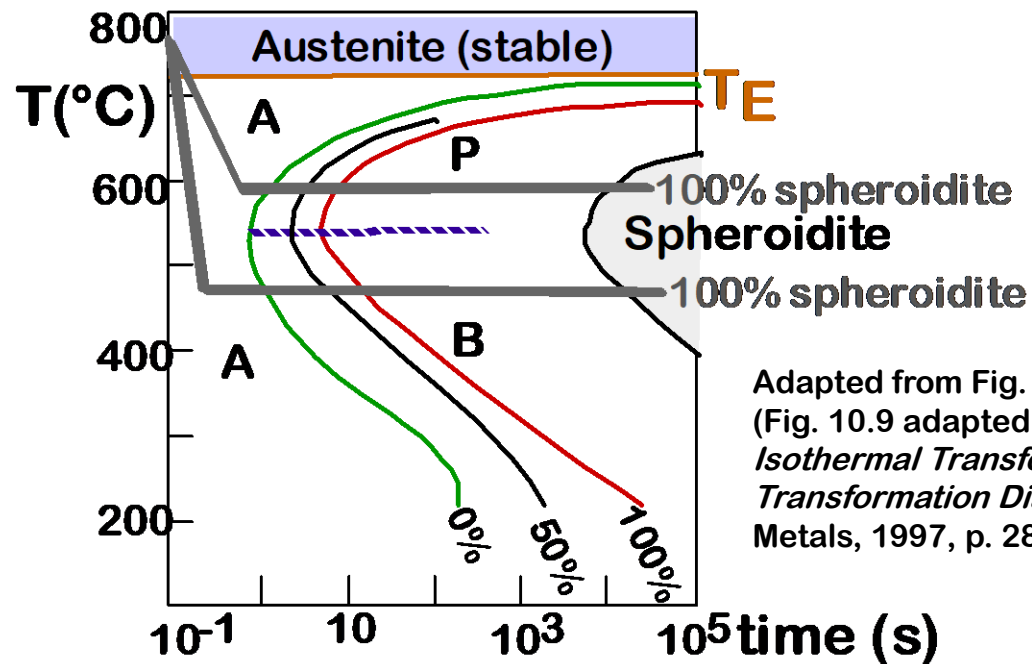
(Adapted from Fig. 10.8, *Callister, 6e*. (Fig. 10.8 from *Metals Handbook*, 8th ed., Vol. 8, *Metallography, Structures, and Phase Diagrams*, American Society for Metals, Materials Park, OH, 1973.)

Bainite reaction rate:

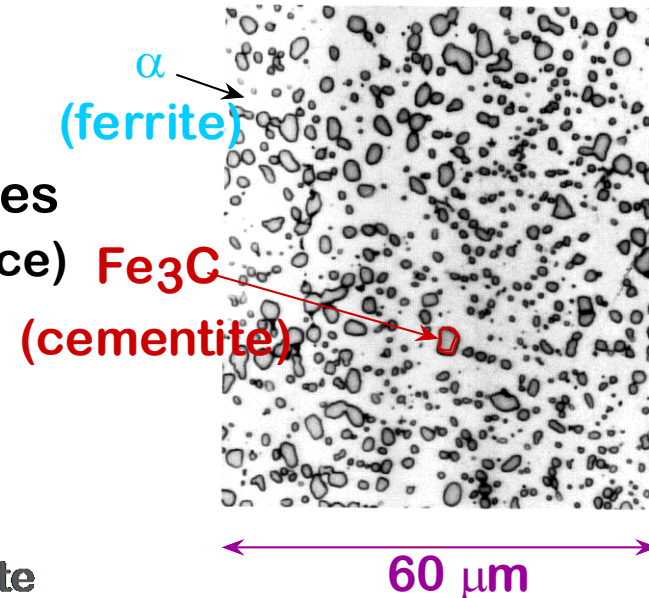
$$r_{\text{bainite}} = e^{-Q/RT}$$

OTHER PRODUCTS: Fe-C SYSTEM (1)

- **Spheroidite:**
 - α crystals with spherical Fe_3C
 - diffusion dependent.
 - heat bainite or pearlite for long times
 - reduces interfacial area (driving force)
- Isothermal Transf. Diagram



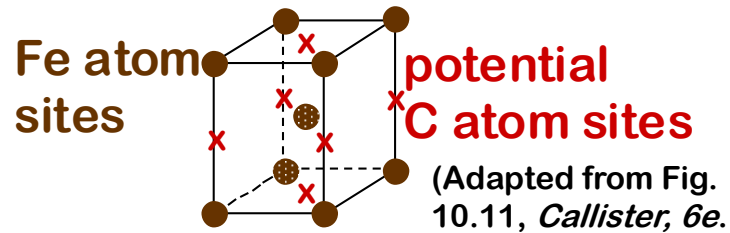
Adapted from Fig. 10.9, *Callister 6e*.
(Fig. 10.9 adapted from H. Boyer (Ed.) *Atlas of Isothermal Transformation and Cooling Transformation Diagrams*, American Society for Metals, 1997, p. 28.)



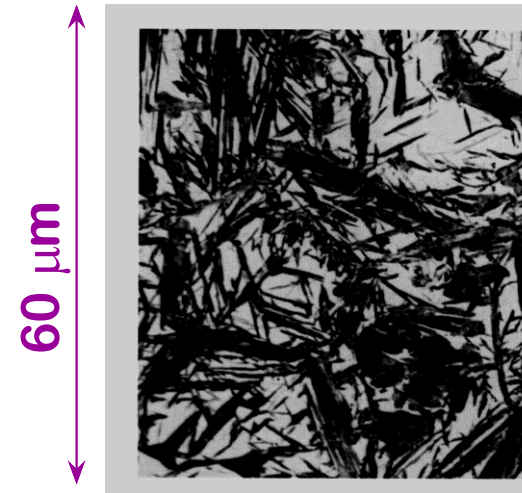
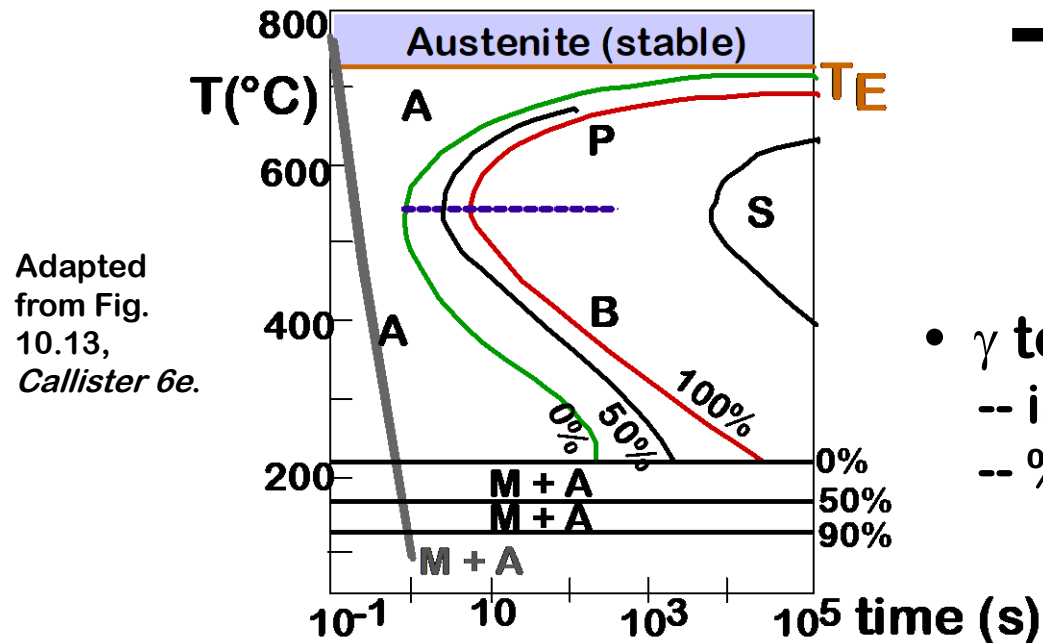
(Adapted from Fig. 10.10, *Callister, 6e*. (Fig. 10.10 copyright United States Steel Corporation, 1971.)

OTHER PRODUCTS: Fe-C SYSTEM (2)

- **Martensite:**
-- γ (FCC) to Martensite (BCT)
(involves single atom jumps)



- **Isothermal Transf. Diagram**



 Martentite needles
 Austenite

(Adapted from Fig. 10.12, *Callister, 6e*. (Fig. 10.12 courtesy United States Steel Corporation.)

- γ to M transformation..
 - is rapid!
 - % transf. depends on T only.

COOLING EX: Fe-C SYSTEM (1)

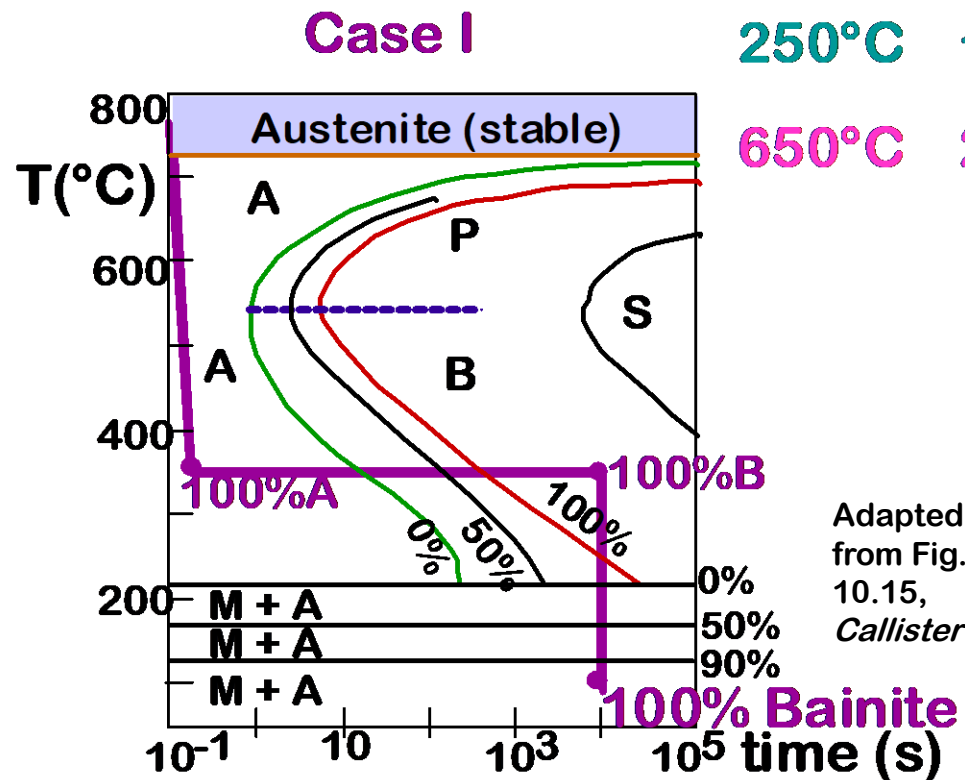
- $C_0 = C_{\text{eutectoid}}$
- Three histories...

Rapid cool to: Hold for: Rapid cool to: Hold for: Rapid cool to:

350°C 10^4 s T_{room} 10^4 s T_{room}

250°C 10^2 s T_{room} 10^2 s T_{room}

650°C 20s 400°C 10^3 s T_{room}



COOLING EX: Fe-C SYSTEM (2)

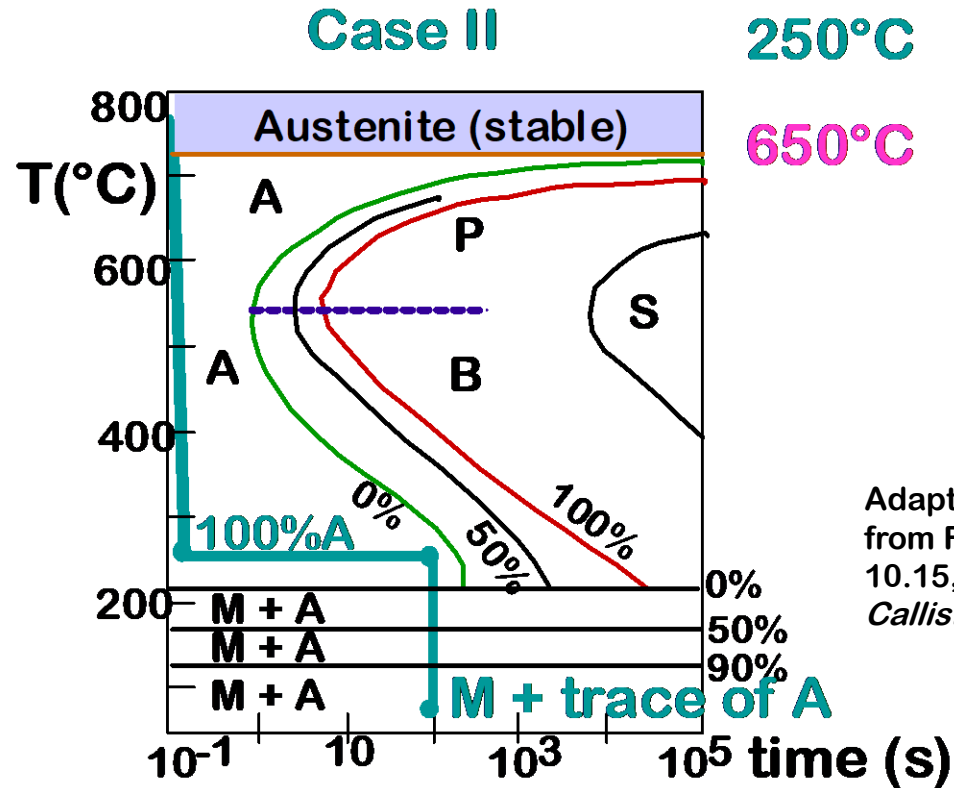
- $C_0 = C_{\text{eutectoid}}$
- Three histories...

Rapid cool to: Hold for: Rapid cool to: Hold for: Rapid cool to:

350°C 10^4 s T_{room} 10^4 s T_{room}

250°C 10^2 s T_{room} 10^2 s T_{room}

650°C 20s 400°C 10^3 s T_{room}



Adapted
from Fig.
10.15,
Callister 6e.

COOLING EX: Fe-C SYSTEM (3)

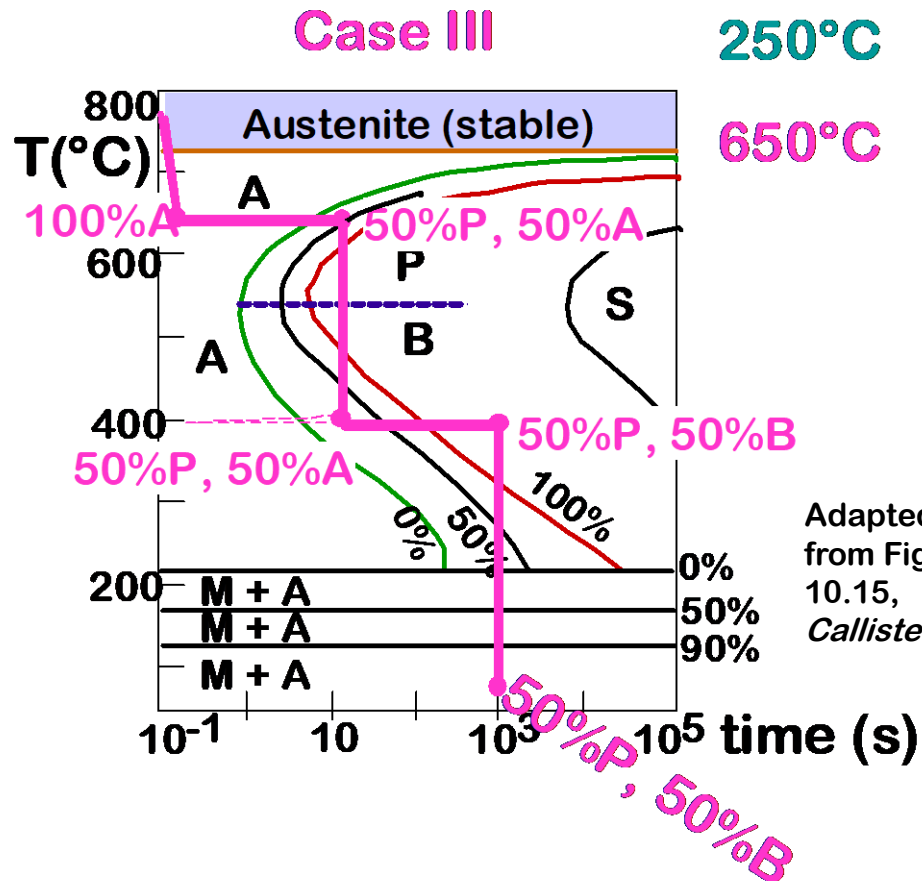
- $C_0 = C_{\text{eutectoid}}$
- Three histories...

Rapid cool to: Hold for: Rapid cool to: Hold for: Rapid cool to:

350°C 10^4 s T_{room} 10^4 s T_{room}

250°C 10^2 s T_{room} 10^2 s T_{room}

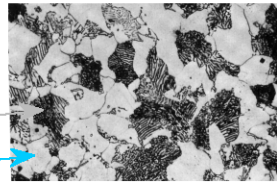
650°C 20s 400°C 10^3 s T_{room}



MECHANICAL PROP: Fe-C SYSTEM (1)

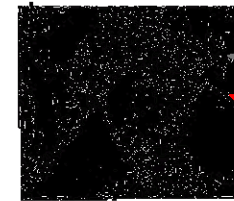
- Effect of wt%C

Pearlite (med)
ferrite (soft)



$C_0 < 0.77 \text{ wt}\% \text{C}$
Hypoeutectoid

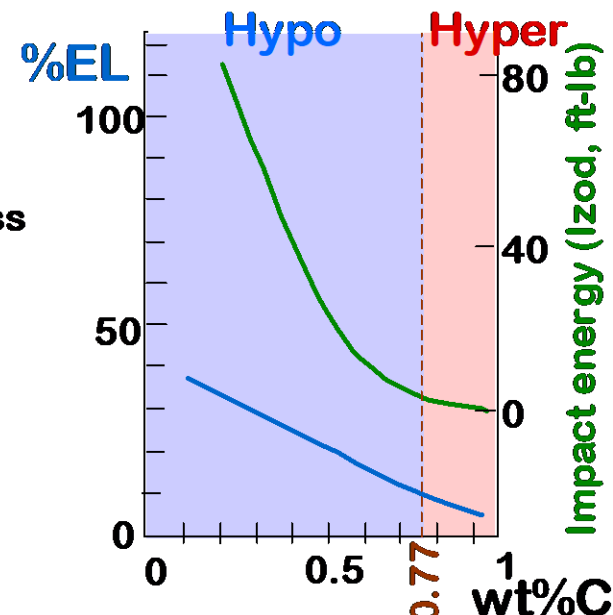
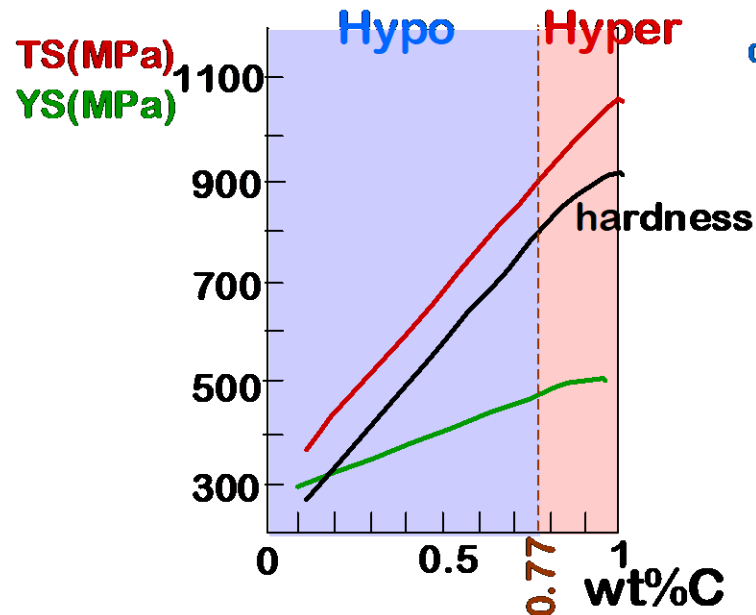
Adapted from Fig. 9.27, *Callister 6e*. (Fig. 9.27 courtesy Republic Steel Corporation.)



Pearlite (med)
Cementite (hard)

$C_0 > 0.77 \text{ wt}\% \text{C}$
Hypereutectoid

Adapted from Fig. 9.30, *Callister 6e*. (Fig. 9.30 copyright 1971 by United States Steel Corporation.)

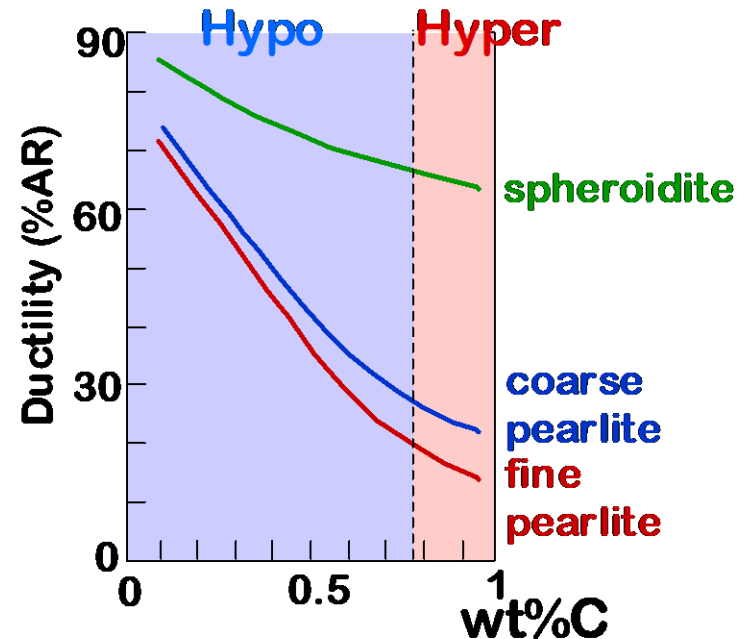
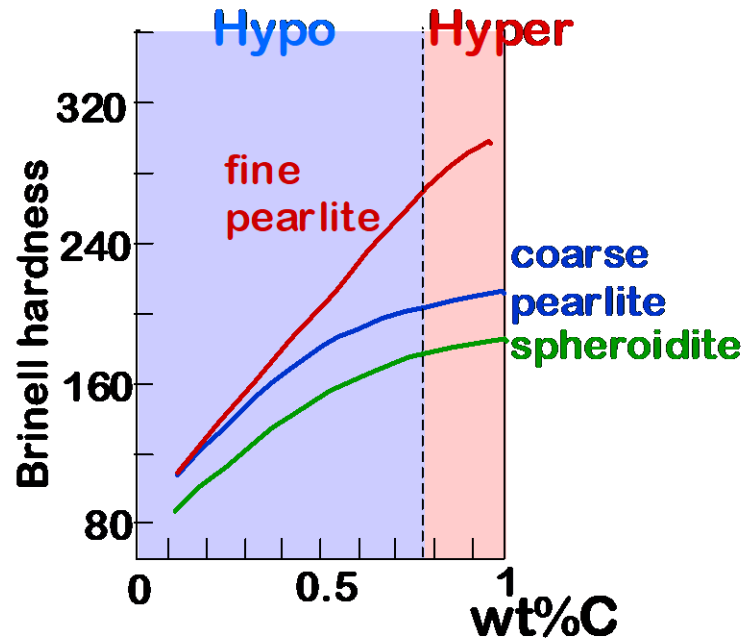


Adapted from Fig. 10.20, *Callister 6e*. (Fig. 10.20 based on data from *Metals Handbook: Heat Treating*, Vol. 4, 9th ed., V. Masseria (Managing Ed.), American Society for Metals, 1981, p. 9.)

- More wt%C: TS and YS increase, %EL decreases.

MECHANICAL PROP: Fe-C SYSTEM (2)

- Fine vs coarse pearlite vs spheroidite

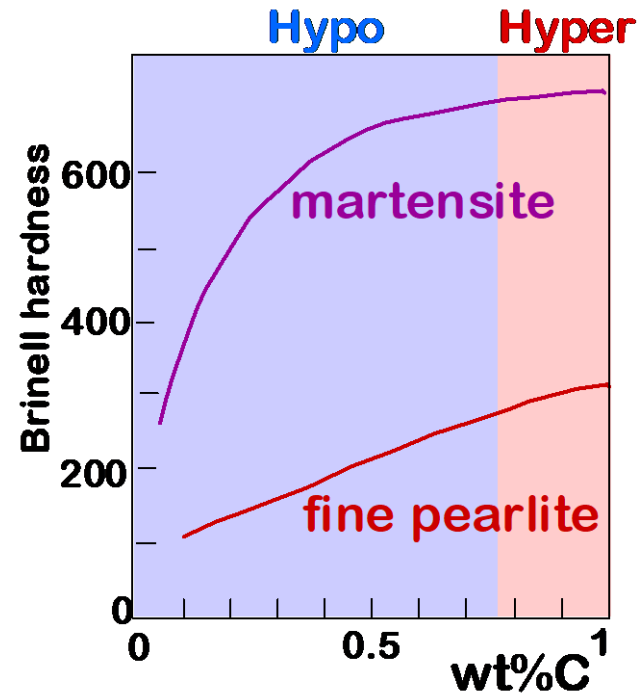


- Hardness: fine > coarse > spheroidite
- %AR: fine < coarse < spheroidite

Adapted from Fig. 10.21, *Callister 6e*. (Fig. 10.21 based on data from *Metals Handbook: Heat Treating*, Vol. 4, 9th ed., V. Masseria (Managing Ed.), American Society for Metals, 1981, pp. 9 and 17.)

MECHANICAL PROP: Fe-C SYSTEM (3)

- Fine Pearlite vs Martensite:

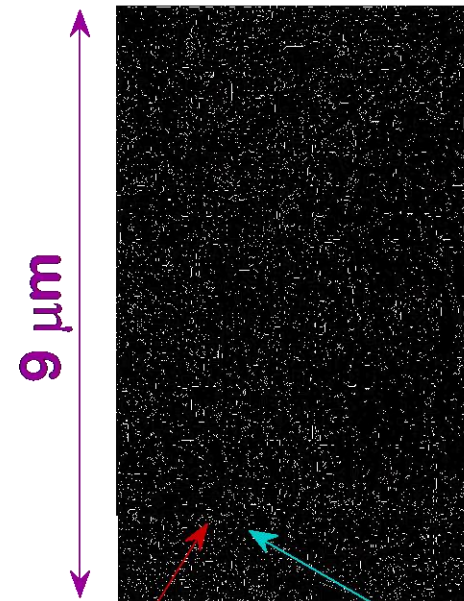
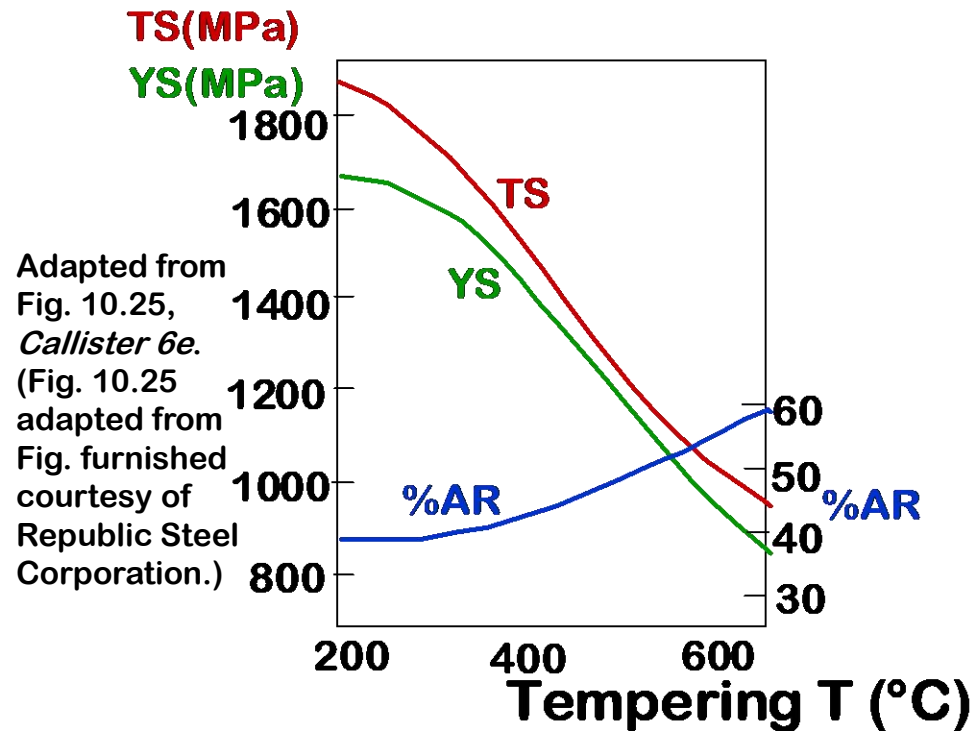


Adapted from Fig. 10.23, *Callister 6e*. (Fig. 10.23 adapted from Edgar C. Bain, *Functions of the Alloying Elements in Steel*, American Society for Metals, 1939, p. 36; and R.A. Grange, C.R. Hribal, and L.F. Porter, *Metall. Trans. A*, Vol. 8A, p. 1776.)

- Hardness: fine pearlite << martensite.

TEMPERING MARTENSITE

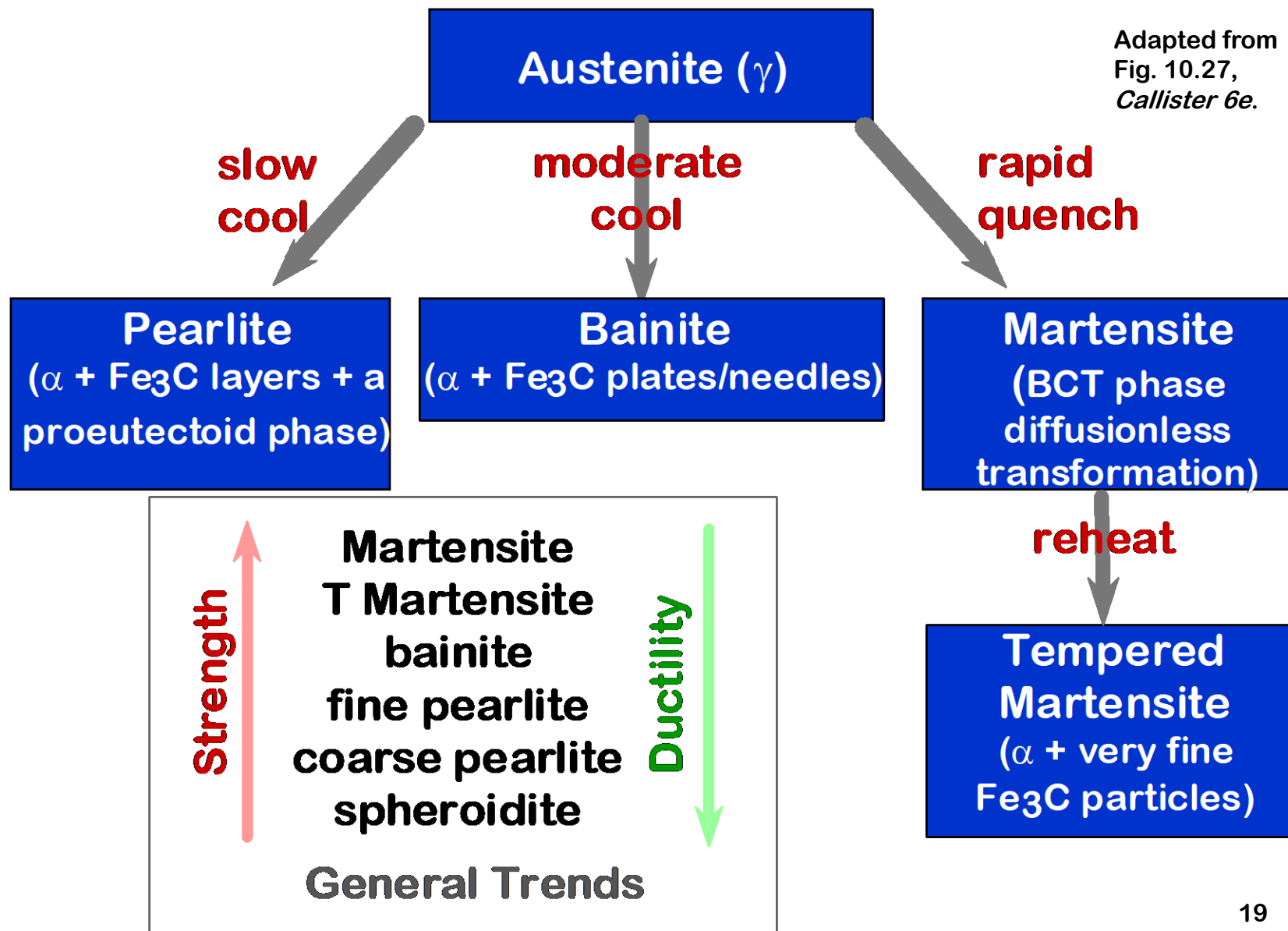
- reduces brittleness of martensite,
- reduces internal stress caused by quenching.



Adapted from Fig. 10.24, *Callister 6e*. (Fig. 10.24 copyright by United States Steel Corporation, 1971.)

- produces extremely small **Fe₃C particles** surrounded by α .
- decreases TS, YS but increases %AR

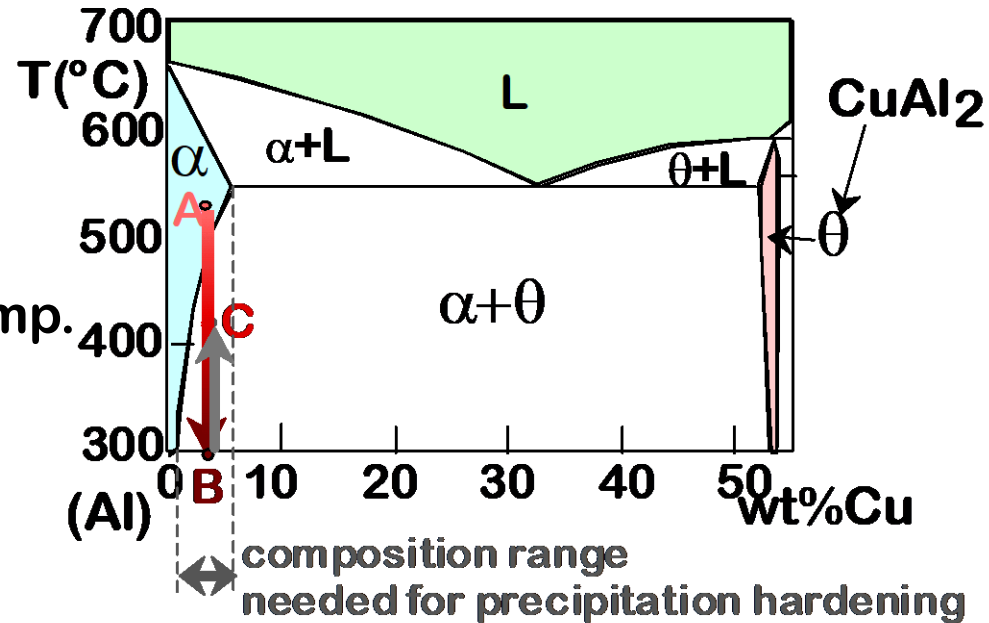
SUMMARY: PROCESSING OPTIONS



PRECIPITATION HARDENING

- Particles impede dislocations.
- Ex: Al-Cu system
- Procedure:

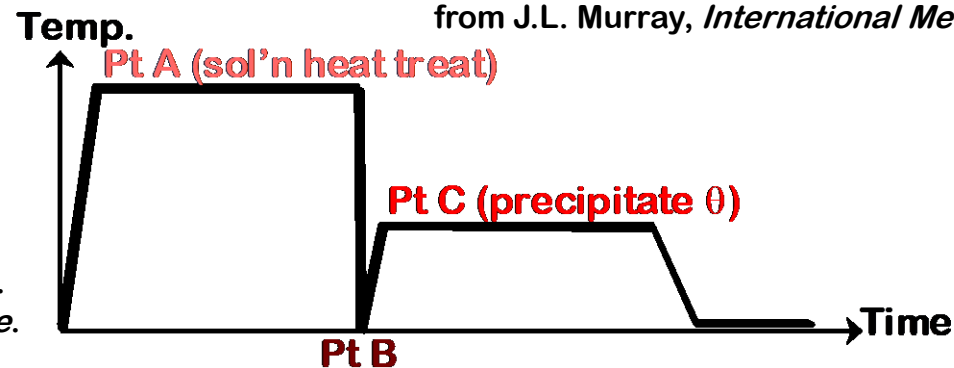
- Pt A: solution heat treat (get α solid solution)
- Pt B: quench to room temp.
- Pt C: reheat to nucleate small θ crystals within α crystals.



Adapted from Fig. 11.22, *Callister 6e*. (Fig. 11.22 adapted from J.L. Murray, *International Metals Review* 30, p.5, 1985.)

- Other precipitation systems:

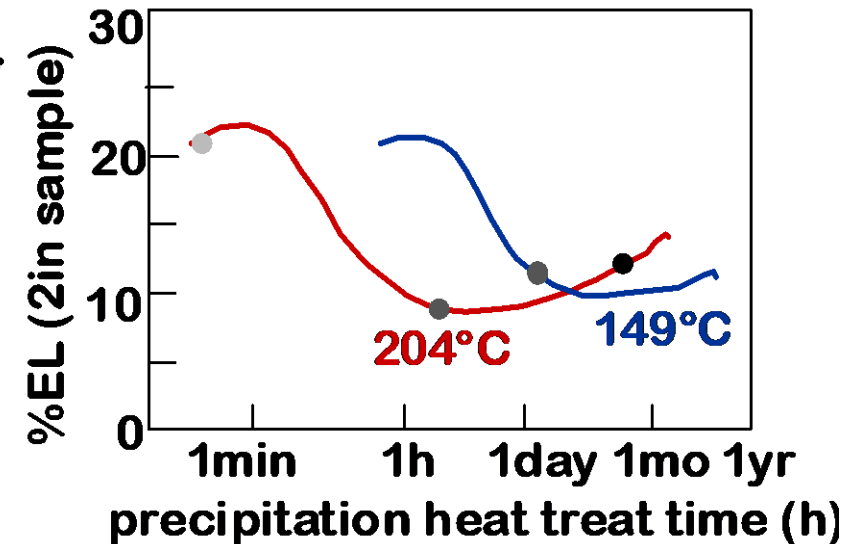
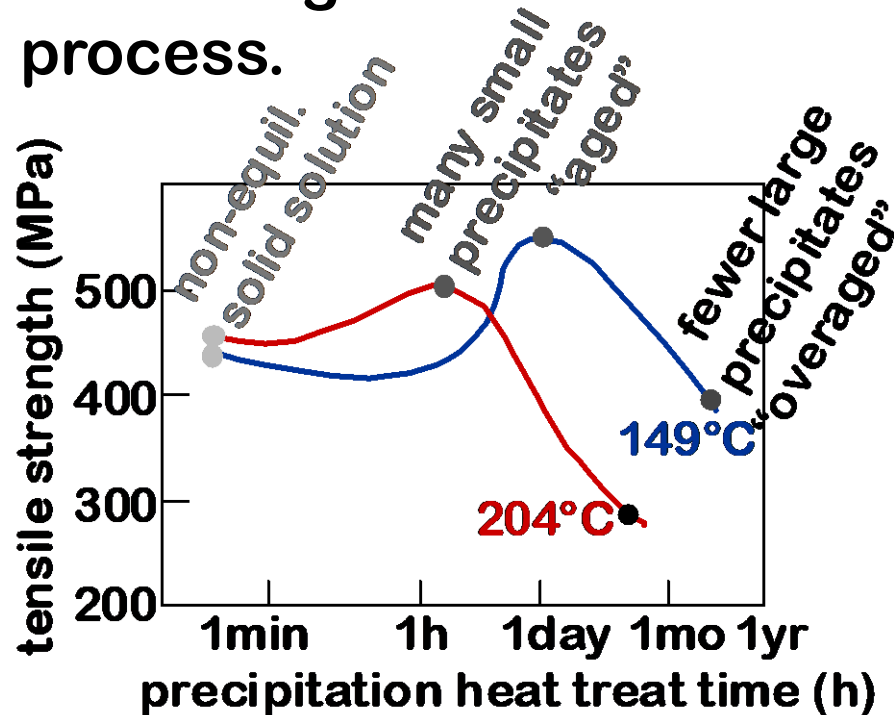
- Cu-Be
- Cu-Sn
- Mg-Al



Adapted from Fig. 11.20, *Callister 6e*.

PRECIPITATE EFFECT ON TS, %EL

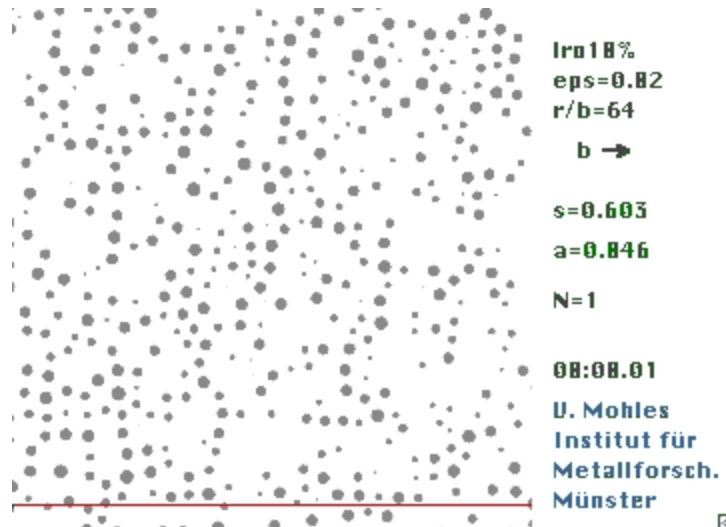
- 2014 Al Alloy:
- TS peaks with precipitation time.
- Increasing T accelerates process.
- %EL reaches minimum with precipitation time.



Adapted from Fig. 11.25 (a) and (b), *Callister 6e*. (Fig. 11.25 adapted from *Metals Handbook: Properties and Selection: Nonferrous Alloys and Pure Metals*, Vol. 2, 9th ed., H. Baker (Managing Ed.), American Society for Metals, 1979. p. 41.)

SIMULATION: DISLOCATION MOTION PEAK AGED MATERIAL

- **Peak-aged**
 - avg. particle size = $64b$
 - closer spaced particles efficiently stop dislocations.

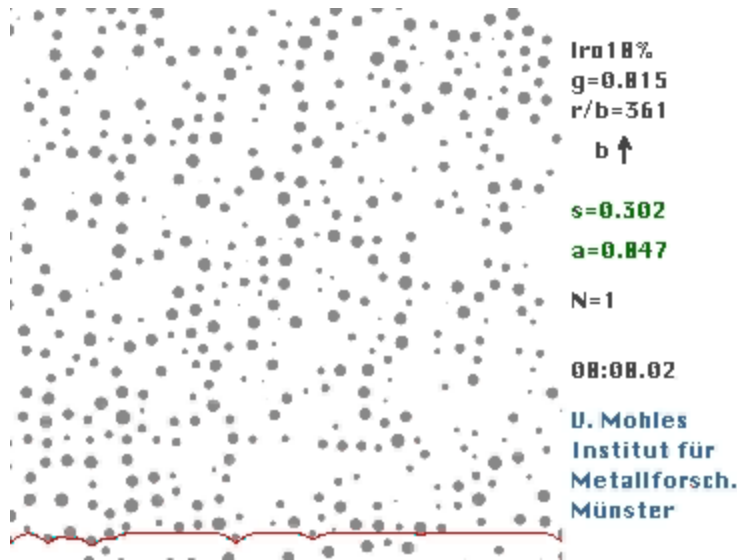


Click on image to
begin simulation

Simulation courtesy
of Volker Mohles,
Institut für Materialphysik der
Universität, Münster, Germany
(<http://www.uni-munster.de/physik/MP/mohles/>). Used with
permission.

SIMULATION: DISLOCATION MOTION OVERAGED MATERIAL

- **Over-aged**
 - avg. particle size = $361b$
 - more widely spaced
 - particles not as effective.

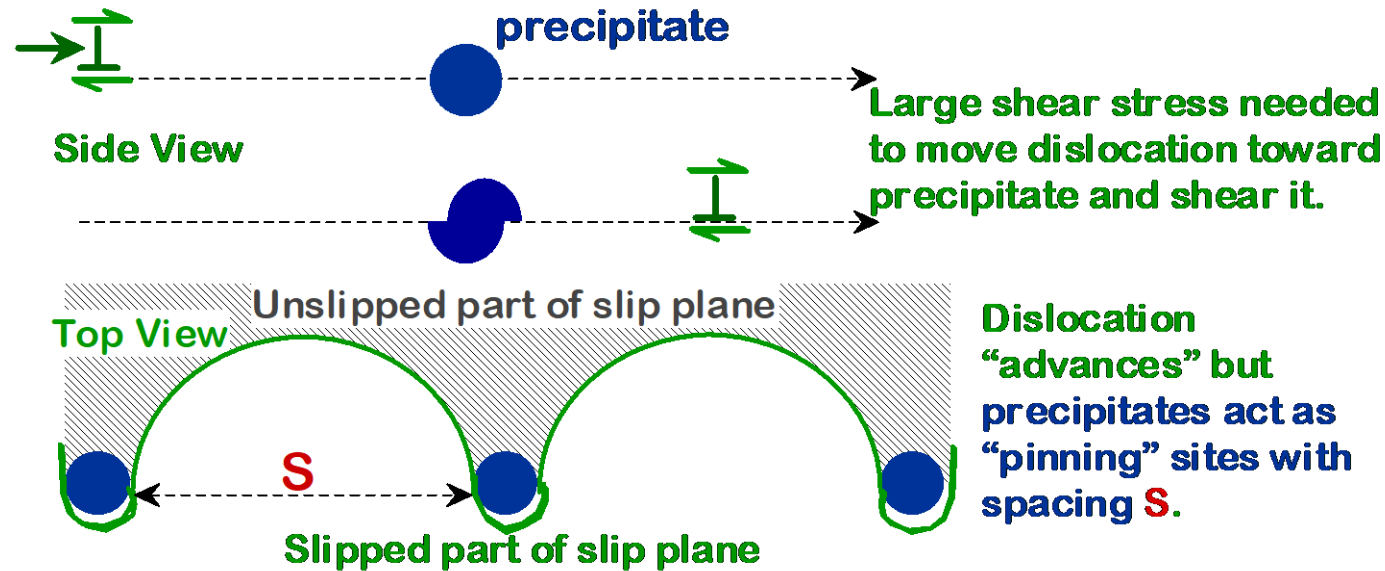


Simulation courtesy
of Volker Mohles,
Institut für Materialphysik der
Universität, Münster, Germany
(<http://www.uni-munster.de/physik/MP/mohles/>). Used with
permission.

Click on image to
begin simulation

STRENGTHENING STRATEGY 3: PRECIPITATION STRENGTHENING

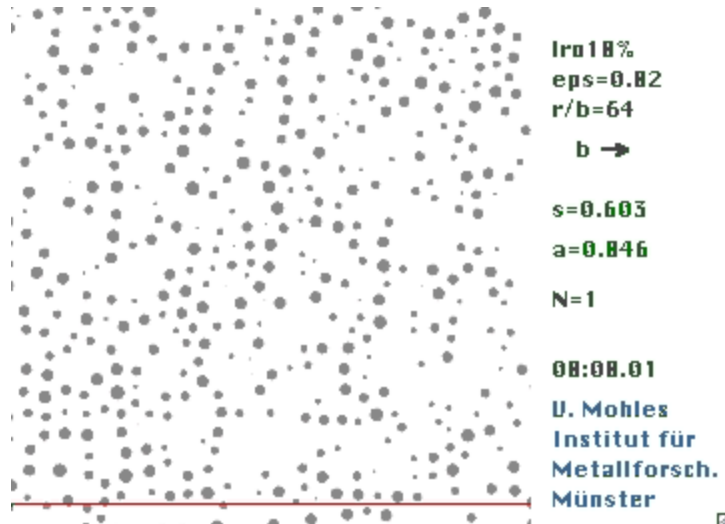
- Hard precipitates are difficult to shear.
Ex: Ceramics in metals (SiC in Iron or Aluminum).



- Result: $\sigma_y \sim \frac{1}{S}$

SIMULATION: PRECIPITATION STRENGTHENING

- View onto slip plane of Nimonic PE16
- Precipitate volume fraction: 10%
- Average precipitate size: 64 b (b = 1 atomic slip distance)



Simulation courtesy of Volker Mohles, Institut für Materialphysik der Universität, Münster, Germany (<http://www.uni-munster.de/physik/MP/mohles/>). Used with permission.

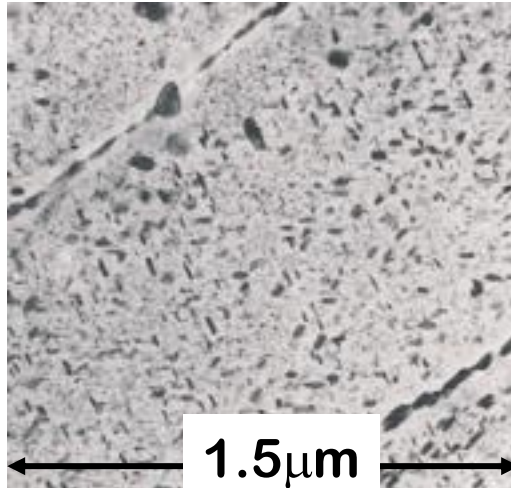
APPLICATION: PRECIPITATION STRENGTHENING

- Internal wing structure on Boeing 767



Adapted from Fig. 11.0, *Callister 5e*.
(Fig. 11.0 is courtesy of G.H. Narayanan and A.G. Miller, Boeing Commercial Airplane Company.)

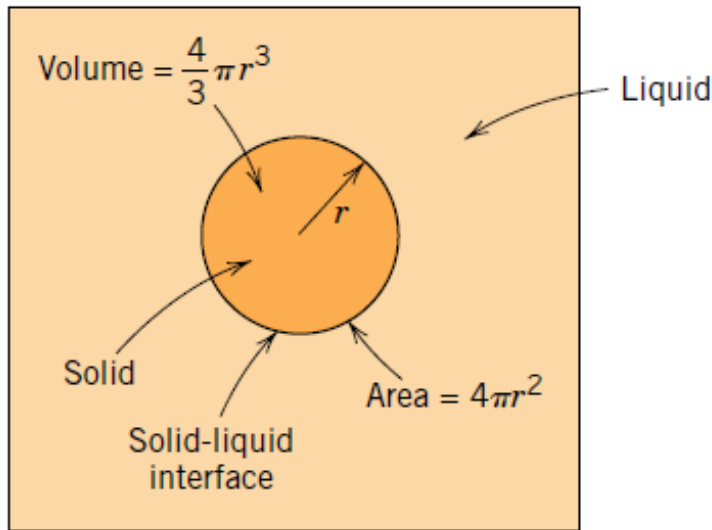
- Aluminum is strengthened with precipitates formed by alloying.



Adapted from Fig. 11.24, *Callister 6e*.
(Fig. 11.24 is courtesy of G.H. Narayanan and A.G. Miller, Boeing Commercial Airplane Company.)

A discussion of the theory of nucleation involves a thermodynamic parameter called **free energy** (or *Gibbs free energy*), G . In brief, free energy is a function of other thermodynamic parameters

A transformation will occur spontaneously only when G has a negative value.



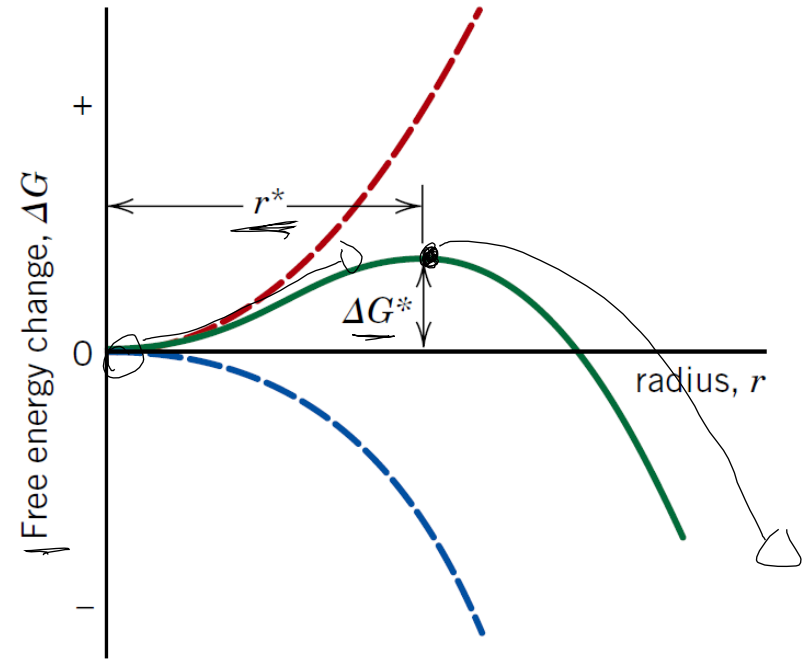
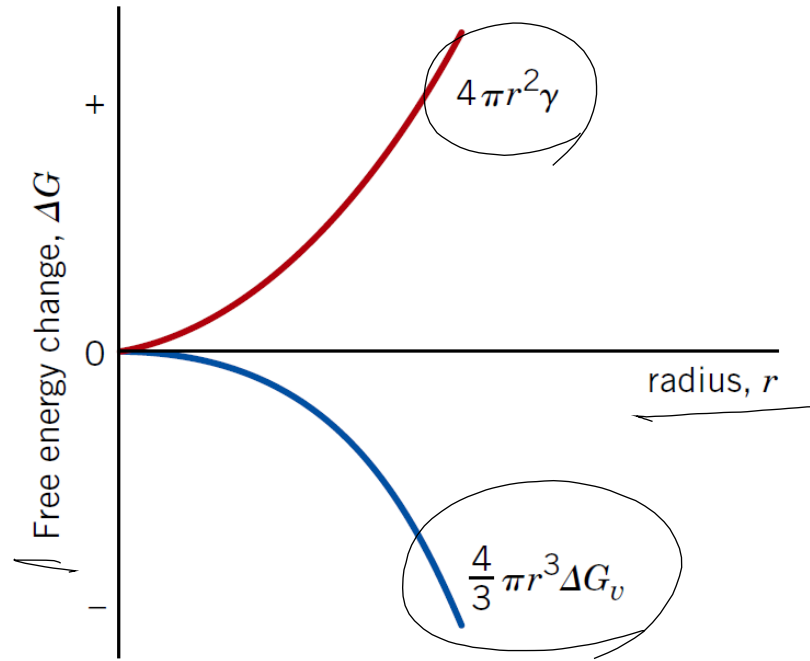
$$\Delta G = \frac{4}{3}\pi r^3 \Delta G_v + 4\pi r^2 \gamma$$

$$\frac{4}{3}\pi r^3 \Delta G_v < 0$$

$$4\pi r^2 \gamma > 0$$

$$\Delta G = \frac{4}{3}\pi r^3 \Delta G_v + 4\pi r^2 \gamma$$

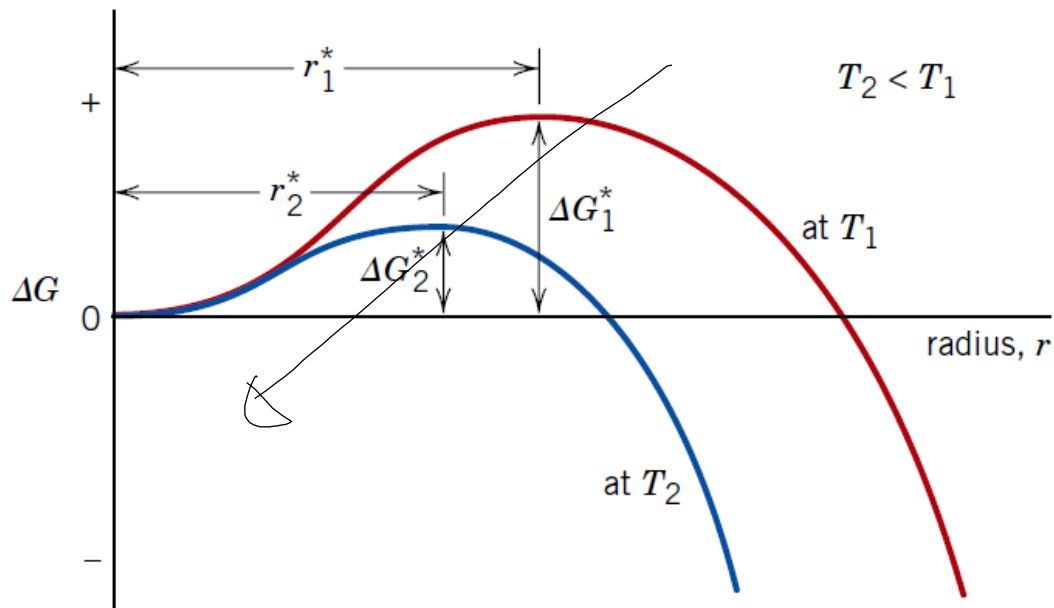
$$\Delta G = \frac{4}{3}\pi r^3 \Delta G_v + 4\pi r^2 \gamma$$



$$* \frac{d(\Delta G)}{dr} = \frac{4}{3} \pi \Delta G_v (3r^2) + 4\pi \gamma (2r) = 0$$

$$r^* = - \frac{2\gamma}{\Delta G_v}$$

$$\Delta G^* = \frac{16\pi \gamma^3}{3 (\Delta G_v)^2}$$

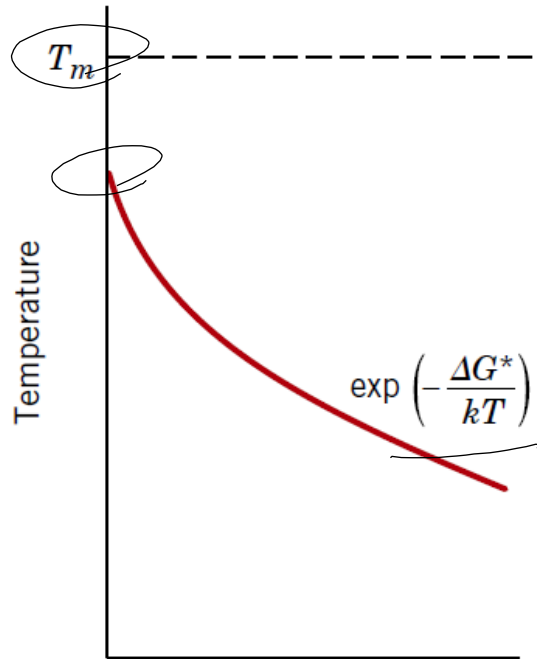


$$\underline{\underline{\delta G_v = \frac{\Delta H_f(T_m - T)}{T_m}}}$$

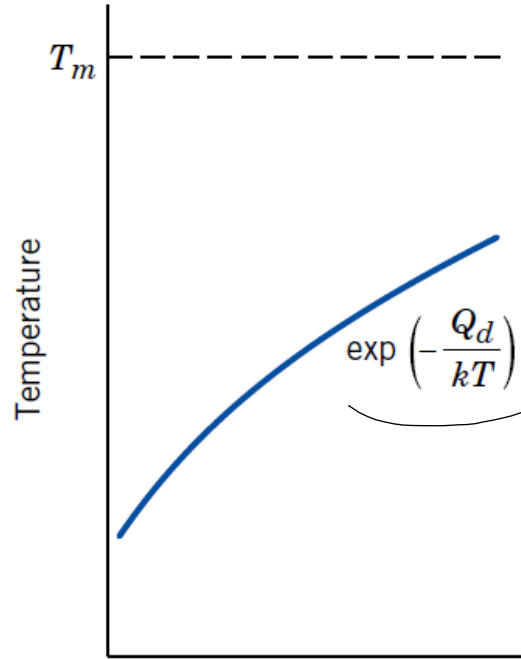
$$r^* = \left(\frac{-2\gamma T_m}{\Delta H_f} \right) \frac{1}{T_m - T}$$

$$\underline{\underline{\Delta G^* = \left(\frac{16\pi\gamma^3 T_m^2}{3\Delta H_f^2} \right) \left(\frac{1}{T_m - T} \right)^2}}$$

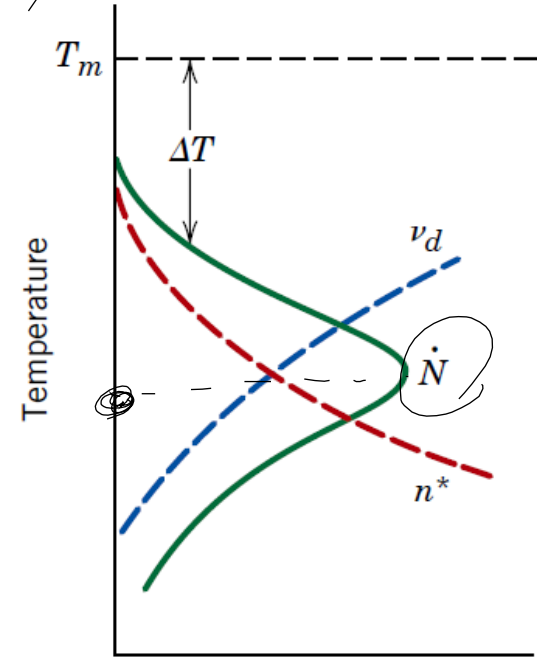
$$n^* = K_1 \exp\left(-\frac{\Delta G^*}{kT}\right) \quad v_d = K_2 \left(-\frac{Q_d}{kT}\right)$$



Number of stable nuclei, n^*



Frequency of attachment, v_d



$n^*, v_d, \dot{N}$

$$w^0 = K_3 n^* v_d = K_1 K_2 K_3 \left[\exp\left(-\frac{\Delta G^* - Q_d}{kT}\right) \right]$$

(a) For the solidification of pure gold, calculate the critical radius r^* and the activation free energy ΔG^* if nucleation is homogeneous. Values for the latent heat of fusion and surface free energy are $-1.16 \times 10^9 \text{ J/m}^3$ and 0.132 J/m^2 , respectively. Use the supercooling value found in Table 10.1.

(b) Now calculate the number of atoms found in a nucleus of critical size. Assume a lattice parameter of 0.413 nm for solid gold at its melting temperature.

$$23.0 \text{ K} \\ = T_m - T$$

$$a) \quad r^* = \left(- \frac{2 \gamma T_m}{\Delta H_f} \right) \left(\frac{1}{T_m - T} \right) = 1.32 \text{ [nm]}$$

$$\Delta G^* = \left(\frac{16 \pi \gamma^2 T_m^2}{3 \Delta H_f^2} \right) \left(\frac{1}{T_m - T} \right)^2 = 2.64 \times 10^{-19} \text{ [J]}$$

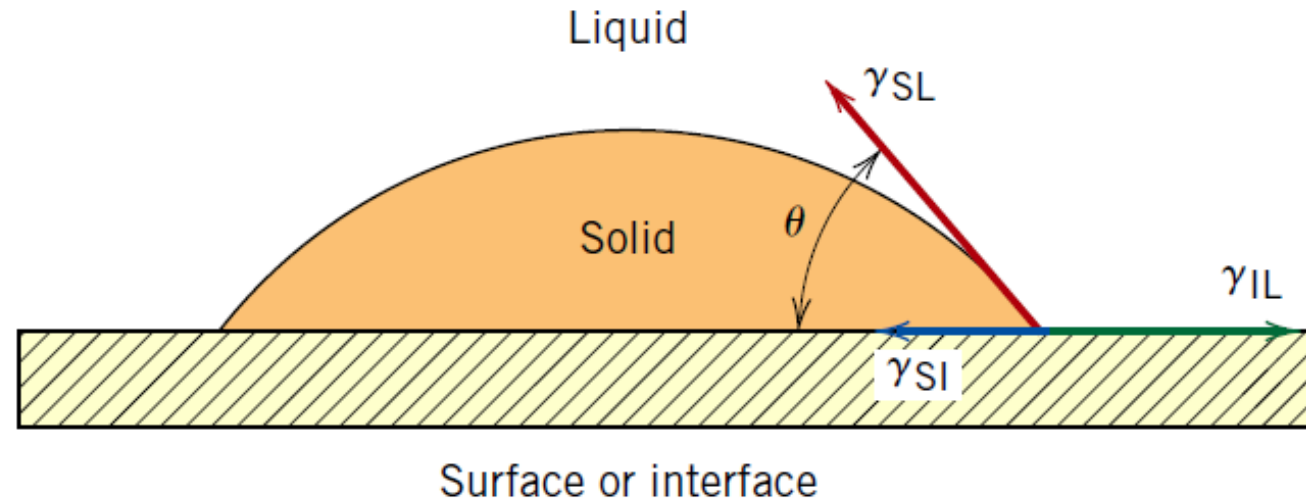
$$b) \quad \# \text{ unit cells} = \frac{\text{volume of nuclei}}{\text{unit cell volume}} = \frac{\frac{4 \pi}{3} r^{*3}}{a^3}$$

$$= 137 \text{ [unit cell]}$$

$$137 \times 4 = 548 \text{ (atoms)}$$

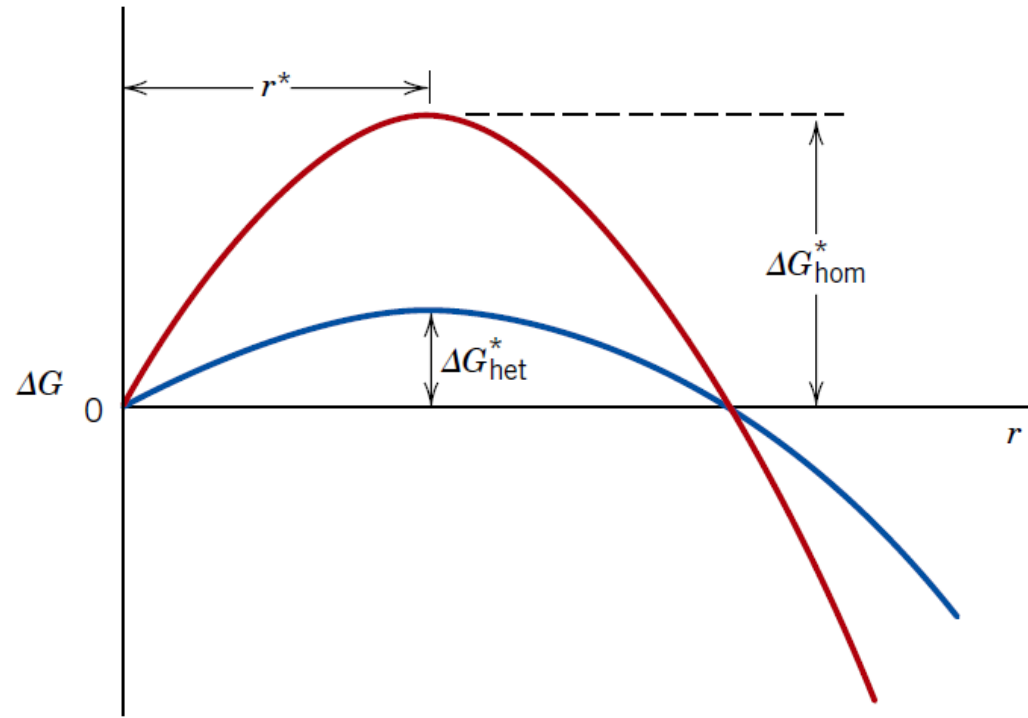
Heterogeneous Nucleation

$$\gamma_{IL} = \gamma_{SI} + \gamma_{SL} \cos \theta$$



$$4\pi r^2 \gamma_{SL}$$

$$\gamma_{IL} = \gamma_{SL} \cos(\theta) + \gamma_{SI}$$



hom
↓

$$0 < S(\theta) < 1$$

$$r^* = \frac{-2\gamma_{SL}}{\Delta G_v} \quad , \quad \Delta G^* = \left(\frac{16\pi\gamma_{SL}^3}{3\Delta G_v^2} \right) S(\theta)$$

$$\Delta G_{het}^* = \Delta G_{hom}^* \times S(\theta)$$